



Droplet jumping stimulated by variation of surface energy: A review from mechanism to application

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ABSTRACT

Droplet deformation alters its surface areas between various phases, leading to changes in its surface energy. When droplet surface area decreases, excess surface energy is released and converted into kinetic energy, enabling droplet jumping. The characteristic velocity of droplet jumping induced by variations in surface energy is typically above 0.1 m/s, and the jumping phenomenon occurs spontaneously without external energy input. The sufficient jumping velocity and spontaneous characteristics make the droplet jumping phenomenon highly promising for widespread applications such as phase change heat transfer, surface anti-icing, anti-frosting, and self-cleaning. This review categorizes droplet jumping into single-droplet and multiple-droplet behaviors, detailing their dynamic characteristics and influencing factors. Both active and passive enhancement strategies are discussed in this study. We summarize practical applications and propose future research directions, offering a comprehensive framework for advancing the understanding and utilization of droplet jumping.

1. Introduction

Droplet jumping has attracted significant attention as an effective mechanism for droplet detachment, with broad potential in applications such as water harvesting [1–3], dropwise condensation [4–6], self-cleaning [7–9], and anti-icing/frosting [10–12]. However, extensive studies have investigated droplet rebound during vertical impact events [13–15], the distinct jumping mechanism—governed by surface energy variation—remains insufficiently explored. Two primary modes of water droplet jumping have been identified: passive droplet jumping (self-jumping) [16,17] and active droplet jumping induced by external energy input [18,19]. Owing to its simplicity and adaptability, self-jumping is particularly promising for practical implementations. This work focuses on the self-jumping phenomenon, which arises from the release and conversion of surface energy during rapid interfacial deformation.

The driving force behind droplet jumping originates from the droplet's intrinsic dynamics and its interaction with the solid-liquid interface, typically facilitated by low-adhesion superhydrophobic surfaces. At micro- and nanoscale, capillary, viscous, and inertial forces govern the complex jumping dynamics. Current research on droplet jumping is

classified into two primary categories. One discusses the dynamic analysis of the process, the mechanism of droplet jumping, and the effects of different parameters. The other outlines how to enhance droplet jumping efficiency. Superhydrophobic surfaces are usually realized by low-energy materials or orderly ordered micro-nanostructures with broad contact angles and modest adhesion forces [20]. The primary objective of enhancing droplet jumping is to investigate innovative designs for practical, durable, and reasonably priced superhydrophobic functional surfaces that encourage droplet jumping and open the door to industrial applications [21–23].

The dynamic behavior of droplets varies across scales, with viscosity, capillary, and inertial forces interplaying to govern jumping characteristics. The initial state of the droplet and the properties of the superhydrophobic surface significantly influence jumping performance, necessitating standardized metrics for evaluation. Two key metrics are commonly used: the dimensionless jumping velocity $v_j^* = v_j/U$ (where v_j is the droplet's jumping velocity and $U = \sqrt{\gamma/\rho R}$ is the capillary scale velocity) and the energy efficiency $\eta = E_k/\Delta E_s$, here E_k is the droplet's kinetic energy of jumping moment and ΔE_s is the excess surface energy that provides the droplet's total energy.

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This review classifies surface energy-driven droplet jumping into single-droplet jumping (driven by Laplace pressure) and multiple-droplet jumping (coalescence-induced). A comprehensive analysis is provided for their respective mechanisms, dynamic behaviors, and enhancement strategies. In addition, practical applications are discussed, and future research directions are proposed to advance the development of droplet jumping (Fig. 1).

Each molecule within the droplet experiences uniform, attractive forces from all directions, resulting in an equilibrium of intermolecular forces. This microscopic equilibrium drives the droplet to transition toward a lower energy state. It is well known that energy input is essential for forming a surface. Assuming the droplet deforms to alter its surface area by dA , the work associated with this deformation is proportional to the number of molecules that must be brought to the surface. Therefore, the energy change can be expressed as:

$$\delta W = \gamma dA \quad (1)$$

Extending the above expression to a macroscopic perspective, the surface energy of the droplet can be expressed as:

$$E_s = \gamma_{lv}A_{lv} + \gamma_{sl}A_{sl} + \gamma_{sv}A_{sv} \quad (2)$$

Where γ is the tension, subscripts s, l, and v denote the solid, liquid, and vapor. As the droplet evolves toward a lower-energy state, the surface energy decreases during specific processes. This reduction in surface energy is referred to as excess surface energy:

$$\Delta E_s = E_1 - E_2 \quad (3)$$

Where E_1 is the initial surface energy of the droplet, and E_2 is the final surface energy of the droplet. This portion of energy may be converted into gravitational potential energy, viscous dissipation, or dissipative ineffective energy such as adhesion work. Importantly, this portion of energy may be converted into upward kinetic energy, thereby inducing droplet jumping. Currently, droplet self-jumping phenomena induced by surface energy variations can be broadly classified into two types:

coalescence-induced jumping and single-droplet jumping.

2. Single-droplet jumping

As the leading research on single-droplet jumping, droplet transportation driven by Laplace pressure difference has been widely studied [24–26]. Vandadi et al. [27] fabricated micro-pillar surfaces with hierarchical roughness and observed that the droplets initially nucleate and grow within the cavities until pinned by surrounding micropillars. Subsequently, the droplets grow vertically out of the plane, eventually transitioning from the Wenzel to the Cassie state through self-stretching driven by the Laplace pressure difference. Although complete single-droplet jumping was not achieved, this work provided valuable insights into the spatial control of condensate droplets. Recent research has shown that the Laplace pressure difference successfully stimulates droplets to jump independently [17,28]. Typically, scholars employ specifically constructed superhydrophobic surfaces (SHSs), such as a groove or cone structure, to induce droplet jumping by Laplace pressure. (Fig. 2a and b) Essentially, single-droplet jumping is a mechanism that utilizes unique structures to cause the conversion of surface energy into kinetic energy.

The jumping process driven by Laplace pressure differences is generally classified into three distinct stages [17,29,30]. Lateral restrictions compel the droplet to grow vertically in the structure during the first growth stage. During the second stage, the droplet grows laterally when its upper section's three-phase contact line reaches the groove's upper edge. As the droplet base becomes increasingly constricted, a significant curvature difference develops between the upper and lower portions, causing a sharp decrease in the contact angle of the upper region. In conical grooves, detailed force models have demonstrated that droplets remain in a quasi-static state during the growth phase [28] with capillary forces driving their slow movement (Fig. 2c). In the third stage, once the curvature-induced pressure difference exceeds the opposing forces of gravity and friction, the droplet is propelled out of the plane.

Evaluating the jump process is more realistic than focusing only on the force at a particular moment, as factors such as droplet morphology, wetting conditions, and internal fluid motion significantly influence the jumping. Accurate understanding of the mechanism is achieved through an energy model. The excess surface energy is the sole source driving the jump, and represents the total available energy for the process. The excess surface energy is partially converted into gravitational potential energy (E_g) due to the variation in the droplet's center-of-mass height during jumping, partially into adhesion energy (E_a) resulting from changes in the wetting state, and partially into viscous dissipation energy (E_{vis}) caused by internal viscous flow. For nanodroplets, the dissipation caused by contact line motion (E_{cl}) and line tension (E_{lt}) is non-negligible. Single-droplet jumping at the nanoscale has not been experimentally observed, and it remains unclear whether contact line friction and line tension are responsible for the failure of droplet jumping. However, in the self-dewetting of nanoscale droplets, contact line friction accounts for the most energy dissipation, and energy analysis based on contact line friction and line tension can inform structural design. In the end, the excess energy is converted into kinetic energy (E_k). The above energy conversion can be expressed using the formula (the exact formula for calculating energy is provided in Table 1):

$$\Delta E_s = E_k + E_{vis} + E_g + E_a + E_{cl} + E_{lt} \quad (4)$$

It is evident that specially designed confinement structures significantly influence droplet jumping. This is because such structures affect the morphological changes of droplets, thereby governing the variation in surface energy.

2.1. Droplet and structure size

Droplet size governs the wetting state and dynamical regime during

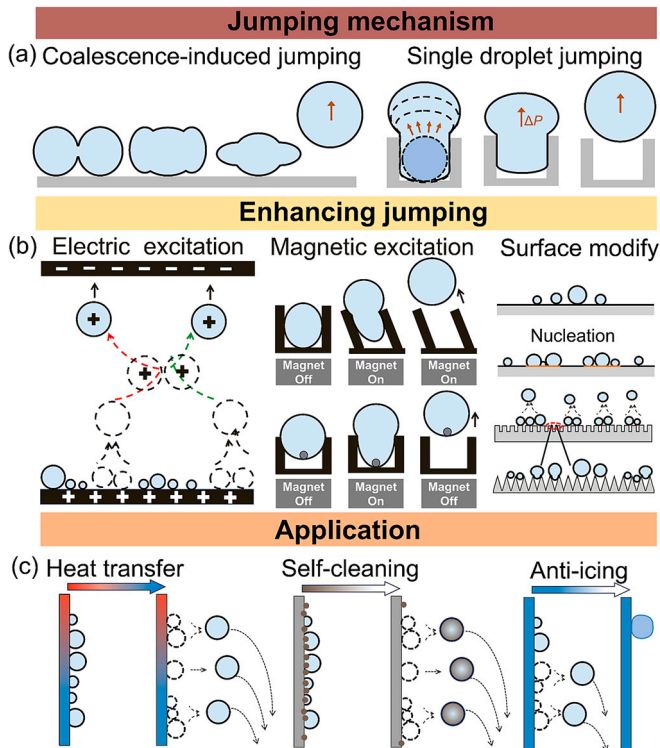


Fig. 1. Self-propelled droplet jumping stimulated by variation of surface. (a) the mechanism of droplet jumping; (b) methods to enhance droplet jumping; (c) the applications of droplet jumping.

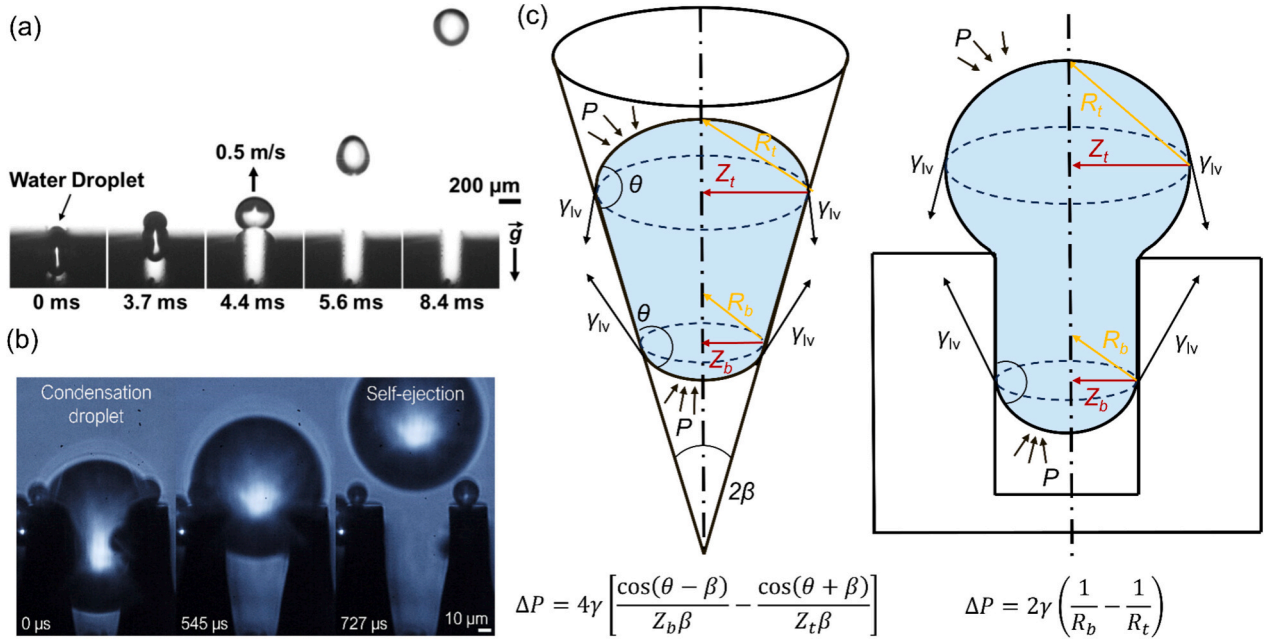


Fig. 2. Single-droplet jumping dynamic analysis. (a) single-droplet jumping in the limited groove [17]. Copyright 2020 American Chemical Society; (b) single-droplet jumping in the limited conical structure [28]. Copyright 2024 American Chemical Society; (c) force analysis model of single-droplet jumping.

Table 1

Specific calculation formula of energy analysis, where A_0 is the initial area of the droplet, A_1 is the area of droplet at the jumping moment, Φ is the dissipation function, A_{sl} is the solid-liquid contact area, Δh is the height difference between the droplet's center of mass in its initial and highest states, μ_{cl} is the contact-line friction coefficient, v_{cl} is the contact-line velocity, L is the characteristic length, τ_{cl} is the contact-line dissipation time, σ_k is the line tension coefficient, and L_{lt} is the total contact line length of single droplet.

Energy	Formula
Surface energy	$\Delta E_s = \gamma(A_0 - A_1)$
Viscous dissipation	$E_{vis} = \int_0^t \int_0^V \Phi dV dt$
Adhesion work	$E_a = A_{sl}(1 + \cos\theta)\gamma$
Gravitational potential	$E_g = \frac{1}{2}\rho V_0 g \Delta h$
Kinetic energy	$E_k = \frac{1}{2}\rho V_0 v_j^2$
Contact line movement	$E_{cl} = \mu_{cl} v_{cl}^2 L \tau_{cl}$
Line tension	$E_{lt} = \sigma_k L_{lt}$

jumping. Unlike coalescence-induced jumping, where the excess surface energy originates from the purely physical coalescence phenomenon, the dynamics of single-droplet jumping are strongly influenced by the interaction between the droplet and the underlying structure. In both scenarios, geometric relationships are essential: the size mismatch between droplets in coalescence-induced jumping, and the size ratio between a single droplet and the structure in single-droplet jumping, both reflect how surface energy variation is governed by geometrical constraints. Therefore, the size-matching [31,32] problem of coalescence jumping can be equivalently interpreted as the droplet-to-structure matching problem of single-droplet jumping. For grooved structures, a compact design is essential to maximize the generation of Laplace pressure by the droplet, implying that the groove can be approximated as a two-dimensional structure. At this point, the groove's geometric parameter (aspect ratio) and the droplet's geometric parameter (radius) can be effectively combined into a single-factor droplet nonsphericity ϕ :

$$\phi = \sqrt{\left(\frac{K^{2P/3} + 2K^{-P/3}}{3}\right)^{1/P} - 1} \quad (5)$$

Where K is the size ratio of the groove, and $P = 1.6$ is derived from the volume of an ellipsoid. Introducing the capillary velocity $\sqrt{\gamma/\rho R}$, the scaling law for the size matching between grooves and droplets is derived as $v_j = \beta\phi\sqrt{\gamma/(0.5WK^{2/3})}$. Combining the eqs. $R = 0.5WK^{2/3}$, the scaling law aligns well with the equivalent size R of the droplets. (Fig. 3a) The situation becomes more complex for structures with inclined sidewalls, such as conical shapes. The curvature introduced by the inclination alters droplet morphology and modifies the Laplace pressure distribution, introducing a dynamic behavior into the analysis [17]. Currently, no comprehensive model quantitatively describes the effect of size mismatch between droplets and conical structures. For jumping phenomena where the droplet and structural dimensions cannot be analyzed together, only the geometric dimensions of the structure or droplet size can be explored independently. Droplet jumping depends on the structure by limiting the droplets' growth to create a pressure differential. The geometric properties of the structure determine how efficiently the droplet jumps. The aspect ratio $K = l/d$, where d is the width of the groove and l is its vertical length, typically illustrates the geometric shape of the structure and represents the structure's ability to constrain droplet morphology. When the aspect ratio K of the rectangular groove grows, the dimensionless droplet jumping velocity increases [29]. When K is small, the aspect ratio rises, further impeding the droplet's ability to develop and increasing the pressure differential that causes the droplet to jump. When K exceeds a certain threshold, the droplet breaks and separates due to Rayleigh-Plateau instability, decreasing its net momentum in the vertical direction [33]. Therefore, identifying a universal aspect ratio K is essential for guiding the structural design of single-droplet jumping, particularly when precise size matching is unattainable. Based on the current experimental and numerical results, an aspect ratio K within the range of 2–4 (Fig. 3b and c) is considered optimal [13,29,33,34]. Yan et al. [17] investigated the relationship between the energy efficiency η , dimensionless jumping velocity v_j^* , and groove aspect ratio K , reporting peak performance at aspect ratio $K = 3$, with dimensionless jumping velocity $v_j^* = 0.95$, and energy efficiency $\eta = 60\%$. In addition to the aspect ratio, the cone angle is another parameter used to characterize the geometric features of the structure. Stendardo et al. [35] analyzed the jumping associated with various cone angles by numerical simulations, determining that an

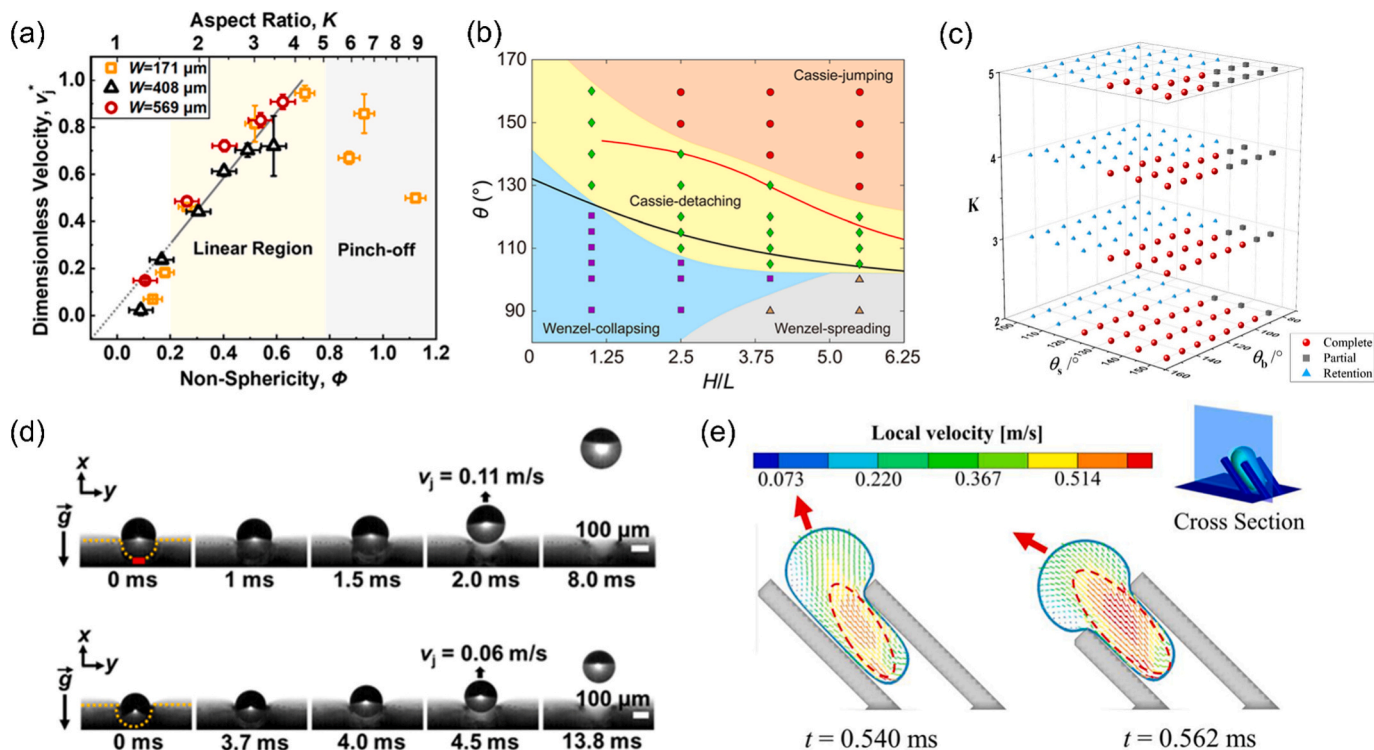


Fig. 3. The influence factor of single-droplet jumping. (a) dimensionless jumping velocity as a function of droplet non-sphericity (Φ) and aspect ratio, representing the influence of size effects on jumping [17]. Copyright 2020 American Chemical Society; (b) single-droplet jumping state diagram [39]. Copyright 2023 Royal Society of Chemical; (c) the impact of contact angle and aspect ratio on single-droplet jumping modes [33]. Copyright 2022 AIP Publishing; (d) the droplet jumping process on biphilic surfaces [17]. Copyright 2020 American Chemical Society; (e) the pressure and velocity vector fields in the asymmetrical groove jumping process [37]; Copyright 2024 Elsevier.

optimal cone angle of 20° maximized jumping performance. It is worth noting that the above conclusions were drawn under conditions where the droplet radius was approximately $200\ \mu\text{m}$ and the structural feature size was around $100\ \mu\text{m}$, indicating a high degree of size matching between the droplet and the structure. This implies that by designing the characteristic dimensions of the structure, the critical size for droplet jumping can be roughly determined. Similar to coalescence jumping, achieving jumping at the micron-scale or even nanometer-scale critical radius remains an interesting and challenging task.

2.2. Hydrophilic points

The driving force behind single-droplet jumping is the Laplace pressure difference, which varies with changes in droplet curvature. Under identical conditions, cylindrical structures typically yield higher jumping velocities than conical structures, as pillar-like geometries are capable of confining larger droplet volumes [35]. Therefore, introducing hydrophilic spots to increase the confined volume of the droplet may serve as an effective strategy to enhance jumping performance. The Namib beetle collects water for survival in arid conditions through the biphilic structure and wettability gradient mechanism on its back [36]. Similar to the water harvesting mechanism of the Namib beetle, single-droplet jumping can also enhance its efficiency by incorporating hydrophilic spots [17,30,35]. Hydrophilic spots facilitate droplet nucleation and create more considerable curvature differences through adhesion, while hydrophobic regions ensure that droplet jumping is not hindered by adhesion. To investigate how the hydrophilic point enhances jumping, Yan et al. [17] conducted experiments and reported that the jumping velocity increased by up to 80 % when hydrophilic spot configuration was employed (Fig. 3d). However, overly large hydrophilic regions hinder droplet jumping by increasing adhesion forces beyond what the Laplace pressure differential can overcome, leading to

pinning. Pu et al. [30] further explored the effect of local hydrophilic sites on droplet jumping by molecular dynamic (MD) simulation. Although significant hydrophilic points can promote droplet growth, the increase in overall pinning force makes droplet jumping more difficult. Despite growing interest in the role of local wettability patterns, a widely accepted dimensionless framework or scaling methodology for quantifying their effects is still lacking. Future research on localized hydrophilic spots will be critical, particularly in establishing a function for jumping velocity concerning hydrophilic spot parameters. Using the model above to guide the design of biphilic surfaces could significantly advance the application of single-droplet jumping in water harvesting or condensation heat transfer.

2.3. Asymmetric groove

An asymmetric pressure distribution is produced when the asymmetric groove upsets the Laplace pressure difference's equilibrium. This imbalance imparts transverse momentum to the droplet, enabling directional motion. For example, asymmetric stepped groove can induce the droplets to jump at a 40° relative to the horizontal. The jumping velocity on asymmetric groove remains comparable to that observed on symmetric grooves, indicating that lateral wall shear has minimal influence on the vertical jumping velocity [17]. Chu et al. [29] conducted numerical simulations to investigate droplet behavior in asymmetric grooves. The droplets accelerated laterally due to the uneven pressure distribution caused by the stepped grooves. Additionally, inclined stepped micropillars create asymmetry flow field of droplet to control the jumping direction. In inclined configurations, the squeezing effect of micropillars on droplet motion depends not only on the direction of the micropillars but also on the expansion direction of the droplet at the top (Fig. 3e). The overall velocity difference between the two is minor. It aligns with experimental findings. However, the jumping angle varies

significantly due to differences in droplet expansion modes [37,38]. Asymmetric groove droplet jumping has excellent promise for directed droplet transport. However, the intricacy of droplet movement within the groove makes accurate directional transport of droplets through asymmetric grooves challenging.

3. Coalescence-induced jumping

In 1969, Kollera and Grigg [40] observed that condensing mercury droplets on a smooth surface spontaneously coalesce and jump. With the advancements in imaging technology, high-speed cameras [41] have been widely applied to droplet research. In 2000, Thoroddsen and Takehara [42] captured the bouncing phenomenon during the coalescence of droplets on liquid surfaces with high-speed camera. Like the phenomena above, when two or more droplets coalesce on a superhydrophobic surface, the coalesced droplet will spontaneously jump. In 2009, Boreyko and Chen [16] discovered and defined this phenomenon as droplet coalescence jumping (Fig. 4a). Coalescence jumping has drawn significant attention due to its potential applications in enhancing heat transfer, self-cleaning, anti-icing, and anti-frosting. The discovery of multi-droplet jumping (coalescence-induced jumping) predates that of single-droplet jumping. The single-droplet jumping was

first observed during experiments investigating groove-assisted droplet coalescence jumping [43]. As a result, research on coalescence jumping is relatively more mature and well-established compared to that on single-droplet jumping.

The droplet coalescence-induced jumping processes are primarily classified using the stages of liquid bridge formation and droplet morphological alterations [44,45]. The first stage involves the formation and growth of the liquid bridge connecting the coalescing droplets. In the second stage, the merged droplet accelerates upward as the liquid bridge collides with the substrate. The third stage is characterized by symmetric contraction of the droplet toward the center and a gradual deceleration. Finally, in the fourth stage, the droplet jumps from the surface (Fig. 4a). The stage classification highlights that both the dynamics of coalescence and the interaction between the liquid bridge and the substrate are critical factors governing the jumping behavior.

The dynamic characteristics of droplet coalescence can be described by length scale and time scale: $l_v = \mu^2/\gamma\rho$, $t_v = \mu^3/\gamma^2\rho$ [46,47]. The magnitudes of the length and time scale determine how the droplet coalescence process evolves. When the coalescence droplet radius R is introduced, the length scale follows the relationship: $l_v/R \sim Oh^2$. The Ohnesorge number (Oh) quantifies the relative influence of viscous, capillary, and inertial forces, and can also be expressed as the ratio of

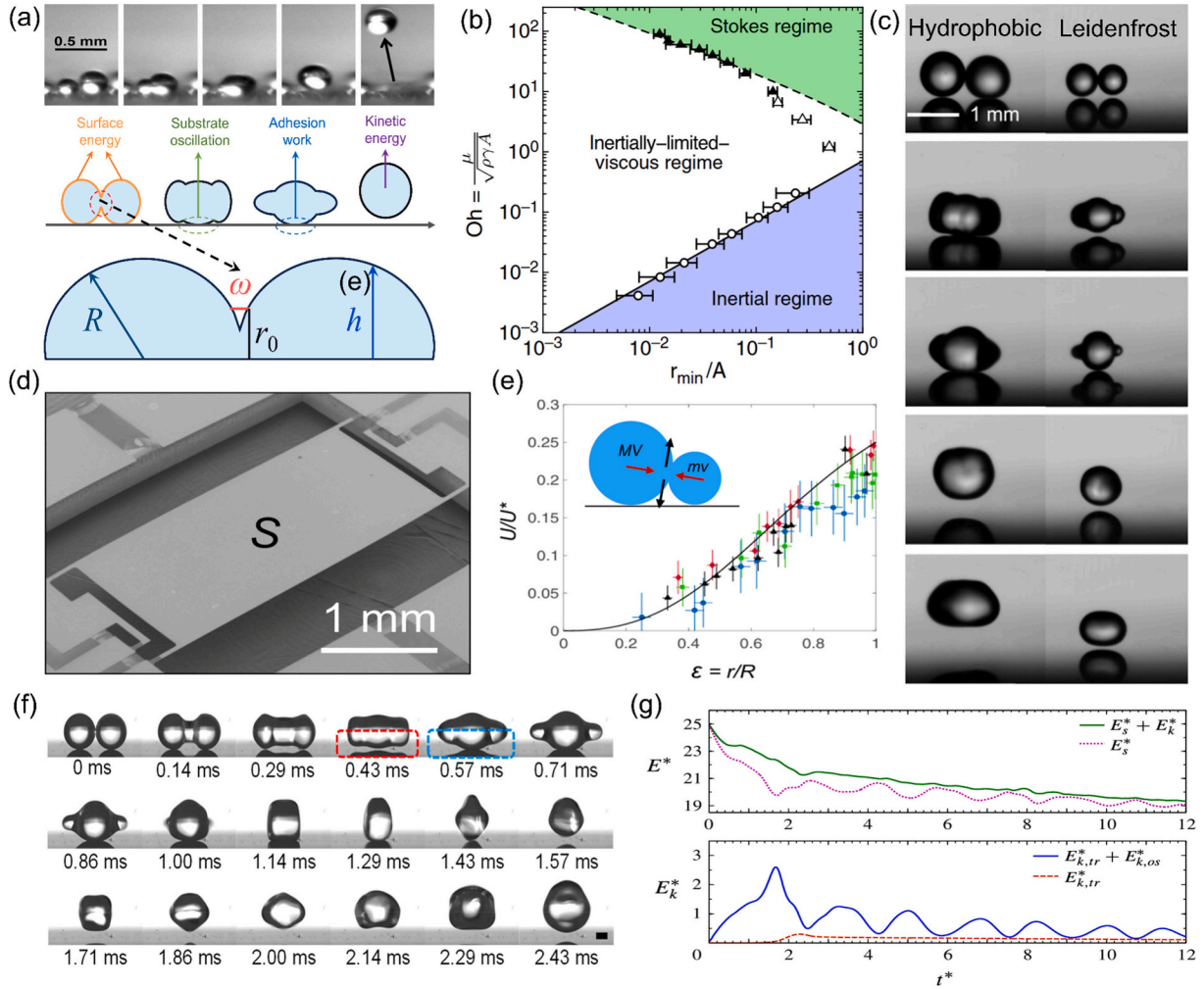


Fig. 4. Coalescence-induced jumping dynamic analysis. (a) schematic diagram of the droplet jumping process and characteristic dimensions [16]. Copyright 2009 American Physical Society; (b) three regimes of droplet coalescence [50]. Copyright 2012 National Academy of Sciences; (c) coalescence-induced jumping in superhydrophobic and Leidenfrost surface [61]. Copyright 2010 AIP Publishing; (d) image of micro sensor for measurement of recoil force; (e) the momentum model for droplet coalescence-induced jumping [62]. Copyright 2017 American Physical Society; (f) coalescence-induced jumping process with regular oscillations [75]. Copyright 2019 American Chemical Society; (g) evolution of the surface energy, translation kinetic energy and oscillatory kinetic energy during the jumping process [76]. Copyright 2014 Cambridge University Press.

characteristic time: $Oh = \tau_v/\tau_i$, where the τ_v and τ_i are the viscous and inertial time scales, respectively. The connection above makes it evident how the three forces—viscous force, capillary force, and inertial force—interact during droplet coalescence (Fig. 4b). At the early stage of coalescence, the curvature difference between the liquid bridge radius R_0 and the droplet radius R is substantial, leading to rapid liquid bridge evolution driven by capillary pressure gradients. [48,49] The temporal evolution of R_0 reflects the dominant force during coalescence, thereby revealing different coalescence mechanisms on a macroscopic scale [50–52]. Hopper [53,54] first derived an exact analytical solution for the initial droplet coalescence in two dimensions, which was later extended to three dimensions by Eggers [55]. The morphological evolution of the liquid bridge is closely linked to the Ohnesorge number (Oh). When $R_0/R < Oh$, coalescence is in an inertial regime dominated by inertial forces [56] and satisfies the relationship $\frac{R_0}{R} \sim \left(\frac{t}{\tau_i}\right)^\alpha$; when $R_0/R > Oh$, coalescence is in a Stokes regime dominated by viscous forces and satisfies the relationship $\frac{R_0}{R} \sim \left(\frac{t}{\tau_v}\right)^\beta$. Recent studies have identified a transitional regime where inertial forces are negligible and capillary forces are balanced by viscous forces. This regime, known as the inertially limited viscous (ILV) regime, is characterized by a unique force balance and coalescence behavior [57,58]. The interaction of capillary, viscous, and inertial forces primarily determines the droplet coalescence state variation. It is not difficult to infer that scaling analysis can be used to link the evolution of the liquid bridge to droplet jumping behavior [59,60]. Differentiate the scaling equation concerning the time given: $v_0 = \frac{1}{2} \frac{CR}{\sqrt{\tau}}$. The failure of coalescence jumping on a Leidenfrost surface indicates that the impact of liquid bridge on the substrate is necessary (Fig. 4c) [61]. Subsequent studies demonstrate that coalescence jumping occurs because the radial evolution of the liquid bridge impacts the surface, breaking the symmetry of the droplet and leading to the jumping. Therefore, the scaling analysis of the radial evolution of the liquid bridge can be regarded as a bridge connecting coalescence dynamics and jumping dynamics. The vertical momentum component driving droplet jumping can be approximated by the scaling of the radial evolution of the liquid bridge. Enright et al. [59] derived the relationship between vertical momentum and the liquid bridge as: $\frac{p_v}{mv} = \frac{3}{2\pi} C^2 \left(\frac{R_0}{R}\right)^3$, here the p_v is vertical momentum of droplet jumping. However, when the droplet radius is less than 100 μm , the predictions of this model gradually deviate from the actual values. Such discrepancies may arise from a shift in the coalescence regime, influenced by droplet size reduction or the presence of surface microstructures. To address the limitations of scaling models, force-sensing platforms fabricated using MEMS technology (Fig. 4d) have been developed to directly measure the momentum generated during the coalescence process [62]. Compared to the scaling model of liquid bridges, sensor measurements are more accurate as they do not need to account for transitions in coalescence dynamics modes. Jumping velocity based on the conservation of momentum can be calculated by:

$$v_j \approx \frac{v_{ci}}{4} = \frac{1}{4} \sqrt{\left(\frac{\gamma}{\rho R}\right)} \quad (6)$$

where v_{ci} is the idealized capillary-inertial velocity. The jumping velocity differs from the capillary-inertial velocity by a factor of 1/4, which is very close to the experimentally measured value $v_j^* \approx 0.20$ [63]. The momentum analysis also matches the experimental data well in cases where the radius of the two droplets differ significantly (Fig. 4e). However, the model overestimates the jumping velocity for smaller microdroplets near the critical size. The analysis based on momentum conservation faces significant challenges due to the difficulty in obtaining the impact force of the liquid bridge and the limitations of scaling the liquid bridge motion. Consequently, energy-based analysis

remains the most widely adopted and reliable method for characterizing droplet jumping behavior [64,65].

3.1. Energy analysis

When two droplets coalesce, surface tension drives the droplet toward a state of lower surface energy. The reduction in solid-liquid interfacial area releases excess surface energy, part of which is converted into kinetic energy and facilitate droplet jumping. Similar to single-droplet jumping, droplet coalescence jumping is also driven by droplet energy conversion through surface area changes. The distinction between the two mechanisms lies in the origin of the surface area change: for droplet coalescence jumping, the surface area variation is achieved through droplet coalescence rather than by restricting droplet growth. Assuming the coalescence of two equal-sized droplets, and neglecting energy losses, the excess surface energy is fully converted into the kinetic energy of the jumping droplet. We can derive the relationship:

$$v_{ji} = \sqrt{\frac{2\Delta E_s}{m}} = 1.11 \sqrt{\left(\frac{\gamma}{\rho R}\right)} \sim \sqrt{\left(\frac{\gamma}{\rho R}\right)} \quad (7)$$

where v_{ji} is the idealized velocity. The energy model and scaling law established a relationship between energy and the capillary scale. Interestingly, when the jumping droplet exceeds the critical size, the dimensionless jumping velocity satisfies the following relationship: $v_j^* \approx 0.2 \sqrt{\gamma/\rho R}$ [66]. Contrary to expectations, only a small fraction of the droplet's surface energy is converted into kinetic energy. The energy efficiency ($\eta = E_k/\Delta E_s$) of coalescence jumping for equal-sized droplets is only 6 % [59]. From an energy perspective, droplet coalescence jumping is a relatively inefficient process. The excess surface energy is not entirely converted into kinetic energy. The majority of the excess surface energy is dissipated through gravitational potential energy, viscous dissipation, and adhesion work, with only a small portion contributing to the kinetic energy that drives jumping [67,68]. The energy analysis of coalescence-induced jumping is represented by Eq. (8).

$$\Delta E_s = E_k + E_{vis} + E_g + E_a \quad (8)$$

Where ΔE_s is the excess surface energy, E_k is the kinetic energy, E_{vis} is the viscous dissipation, E_g is the gravitational potential energy, and E_a is the adhesion work.

Typical droplet coalescence jumping phenomena can be recorded using high-velocity cameras [69,70], where the droplet size ranges from tens to hundreds of microns. At this scale, the droplet radius is much smaller than the capillary length, rendering the influence of gravitational potential energy negligible. Additionally, the influence of the three-phase contact line becomes insignificant. The work of adhesion can be calculated by $E_a = \phi\gamma(1 + \cos\theta)A$. Here ϕ represents the fraction of a Cassie droplet's base in direct contact with the surface structure and A represents the solid-liquid contact area, which can be calculated by $A = (R\sin\theta)^2$ [71,72]. We can roughly estimate the adhesion work. The results indicate that the adhesion work is approximately two orders of magnitude smaller than the excess surface energy. Therefore, the remaining parts (viscous dissipation) may be the reason for the inefficiency of droplet jumping. During the droplet coalescence process, viscous dissipation arises from the internal fluid flow, the formation of the liquid bridge, and deformation of the droplet shape. The complex of these coupled processes makes it challenging to quantify viscous dissipation accurately. Viscous dissipation can be calculated by $E_{vis} = \int_0^\tau \int_0^V \Phi dV dt$ [73,74], where V is the volume of the droplet, τ is the coalescence-induced jumping time, and Φ is the dissipation function. The general form of the dissipation function is derived from the Navier-Stokes equations:

$$\Phi = 2\mu \left\{ \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 + \frac{1}{2} \left[\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 \right] \right\} \quad (9)$$

where u , v , and w represent the velocity components of the droplet along the x -, y -, and z -axis, respectively. The dissipation function typically represents the power loss due to viscosity and is closely related to the fluid's flow characteristics, viscosity, and deformation properties. It provides an effective framework for evaluating viscous dissipation in numerical simulations. However, in the scaling analysis, various factors introduce substantial complexity. Most of the flow is directed along the x -axis during the droplet coalescence. As a result, the dissipation function of the two coalescing droplets enables a clever simplification $\Phi = \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 \approx \mu \left(\frac{U_c}{R} \right)^2$ [74], where U_c is the coalescence velocity. Subsequent studies have also found that most of the viscous dissipation occurs during the droplet coalescence process, confirming the accuracy of the above equation. The calculation results show that the viscous dissipation during the droplet jumping process accounts for 40 % to 60 % of the excess surface energy. However, the total accumulated energy accounts for only 50 % to 70 % of the excess surface energy. This energy deficit suggests that additional dissipation mechanisms may have been overlooked. This result only suggests that surface viscous dissipation is a partial cause of the inefficiency in droplet jumping. When the Ohnesorge number exceeds 1, viscous forces dominate, corresponding to a droplet radius of 10 nm. However, the observed critical radius for droplet jumping is around 10 μm , much larger than the nanometer scale. The significant size difference implies that other factors influence droplet jumping and viscous dissipation.

Experimental observations reveal regular oscillations accompany the droplet jumping process (Fig. 4f) [75]. In other words, the excess surface energy is converted into oscillatory and translational kinetic energy [76]. The translational component contributes directly to the jumping motion, while the oscillatory component varies over time and is reflected in periodic fluctuations of surface energy. At the moment of jumping, the oscillatory kinetic energy is several times greater than the translational kinetic energy. Over time, the oscillatory kinetic energy gradually decreases due to the internal flow of the droplet. The oscillatory kinetic energy follows a sinusoidal function that gradually decreases with time (Fig. 4g). Therefore, the primary causes of the inefficiency in droplet jumping are viscous dissipation and oscillatory kinetic energy. Under ideal conditions, the energy equation for coalescing droplet jumping can be simplified as:

$$\Delta E_s = E_{k, \text{tr}} + E_{k, \text{os}} + E_{\text{vis}} \quad (10)$$

where $E_{k, \text{tr}}$ is the translational kinetic energy, and $E_{k, \text{os}}$ is the oscillatory kinetic energy. Notably, droplet jumping typically occurs in the range of 10 μm to 1000 μm on superhydrophobic surfaces. Therefore, the above analysis is based on ideal and general conditions. If jumping occurs on hydrophilic surfaces, the adhesive work becomes the dominant energy dissipation. When the Bond number ($Bo = \Delta\rho g R^2 / \gamma$), which characterizes the relative importance of gravity to surface tension, is significant and gravitational forces become dominant. Overcoming the variation in gravitational potential becomes the dominant form of energy dissipation. However, the fundamental physical model still applies.

3.2. Critical size

Since the discovery of the droplet coalescence-induced jumping, the influence of the droplet radius on the process has been investigated. Boreyko and Chen initially assumed that all excess surface energy was converted to kinetic energy, leading to the following theoretical relationship between droplet radius and jumping velocity [16]:

$$v_j \sim \sqrt{\gamma / \rho R} \quad (11)$$

This scaling predicts that the jumping velocity should decrease with increasing droplet radius. However, no monotonic relationship is found experimentally between the droplet jumping velocity and droplet radius. Instead, the relationship is contrary to theoretical predictions at the micron scale. This deviation is attributed to a shift in the dominant forces at different regimes. As a result, Boreyko and Chen proposed that the jumping behavior could be divided into two regimes. When the droplet radius is less than 50 μm and the Re number is less than 10, the viscous force dominates, leading to significant viscous dissipation and disruption of the radius-velocity scaling. Conversely, when the droplet radius is greater than 50 μm and the Re number is approximately more significant than 10, the jumping velocity is proportional to the capillary velocity. The critical radius R_{cr} separating these two regimes is defined as:

$$R_{cr} \sim \mu^2 / \gamma \rho \quad (12)$$

The critical radius of a water droplet is determined by the formula above to be 10 nm, but experimental observation indicates that the critical radius is 10 μm . The droplet jump mechanism is not accurately represented by the equation above. Wang et al. [77] developed a refined energy model incorporating viscous dissipation, leading to a modified expression. As a result, the computation of viscous dissipation was simplified:

$$E_{\text{vis}} = 36\pi\mu\sqrt{\gamma R^3 / \rho} \quad (13)$$

The expression is consistent with experimental findings, which indicate that the jumping velocity exhibits a parabolic relationship with the droplet radius. The reason is that below a certain radius, viscous forces dominate over inertial and capillary forces, resulting in energy losses [78,79]. This regime shift is well characterized by the Ohnesorge number ($\mu / \sqrt{\gamma \rho R}$), which characterizes the relative importance of viscous force to inertial and capillary forces. Liu et al. [76] used the Oh number to classify jumping regimes. When $Oh > 0.1$, the dimensionless jumping velocity drops rapidly due to dominant viscous effects. In contrast, when $Oh < 0.1$, the dimensionless jumping velocity remains nearly constant at 0.2. Consequently, most of the subsequent studies use the Oh number instead of the Re number to investigate the effect of droplet viscosity on droplet jumping.

For droplets with smaller radius, viscous dissipation prevails when $Oh > 0.1$, and the excess surface energy generated is insufficient to propel droplet jumping. The dimensionless jumping velocity remains around 0.2 for droplets in the inertial region [75,76,80,81]. The critical size of droplet jumping is closely related to the substrate where droplets coalesce. Merely 10 μm is the critical size for coalescence jumping on a flat superhydrophobic surface (Fig. 5). Substrate modification can significantly reduce this threshold and enhance jumping performance. Kim et al. [82] conducted coalescence jumping experiment on the copper oxide surface and observed successful jumping for droplets smaller than 10 μm , suggesting that droplet-substrate interactions play a critical role in the jumping dynamics. Further studies by Cha et al. [83] demonstrated submicron-scale droplet jumping on superhydrophobic surfaces modified with carbon nanotubes, detecting jumping droplets with radius as small as 500 nm. As research techniques advance, the minimum achievable jumping scale continues to decrease.

In summary, the capacity of droplets to jump rises initially around 10 μm but diminishes as droplet size increases. A smaller critical radius implies a higher droplet refresh rate and nucleation rate, which in turn reduce thermal resistance and enhance heat transfer coefficients in condensation applications. Therefore, minimizing the critical radius is key to improving the practical performance of droplet jumping. To achieve this, structural surfaces should be designed to reduce the critical radius and promote efficient energy conversion.

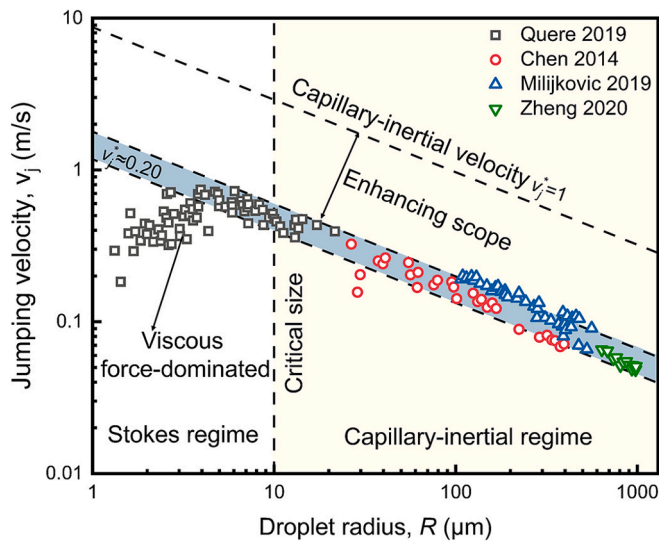


Fig. 5. The measured velocity of jumping droplets as a function of droplet radius is plotted, with data from various studies overlaid on the graph. The legend indicates the corresponding authors of the respective studies [75,76,80,81]. The droplet velocity follows the inertial-capillary scaling law, with the dimensionless velocity approximately equal to 0.2. However, at the critical radius, the influence of viscous forces becomes dominant, causing the velocity to diverge and sharply drop to 0.

3.3. Size mismatch

The jumping of two same-size droplets has been extensively studied. However, it is more typical for droplets of uneven sizes to coalesce in applications. Due to the size difference between the droplets, coalescence occurs in an asymmetric state, leading to asymmetric internal flow within the droplets [31,84]. The recoil force generated by the liquid bridge impact on the substrate reduces, thereby lowering the jumping velocity. When the size difference between droplets becomes too large, the liquid bridge fails to impact the substrate during coalescence, leading to a non-jumping coalescence. Influenced by the asymmetric internal flow, the oscillatory motion of the droplet is disrupted and subsequently converted into rotational motion. Consequently, asymmetric morphological development of the coalesced droplet alters the growth and expansion behavior of the liquid bridge, reducing jumping efficiency. The dominant mechanism behind the inhibition of jumping under size mismatch lies in the weakening of the liquid bridge impact. The jumping velocity declines with the mismatch ratio, here the mismatch is the ratio of tiny droplets to large droplets. Droplets of varying sizes must meet a threshold size mismatch ratio to jump. Experimental studies have shown that a threshold mismatch ratio must be met for successful jumping. When this ratio falls below 0.56, the kinetic energy of the coalescence droplet becomes insufficient to overcome adhesion and lift off the surface (Fig. 6). Xie et al. [85] defined the dimensionless parameters related to the coalescence of droplets with unequal sizes $Oh^* = Oh\sqrt{\frac{2}{1+r^2}}$, where r^* is the ratio of the radius of the two droplets. The scaling law was validated through MD simulations. The results confirm that, within a specific range of Oh^* , the coalescence-induced jumping of unequal-sized droplets closely follows the scaling law, with the dimensionless jumping velocity remaining constant at approximately 0.127. Furthermore, detailed analysis of the coalescence dynamics shows that non-jumping behavior in high mismatch coalescence is primarily due to the failure of the liquid bridge to impact the substrate. This is strongly supported by experimental observations. Although there is broad consensus that droplet size mismatch significantly reduces jumping velocity and energy efficiency, its influence on jumping direction is less understood. Peng et al. [43] conducted

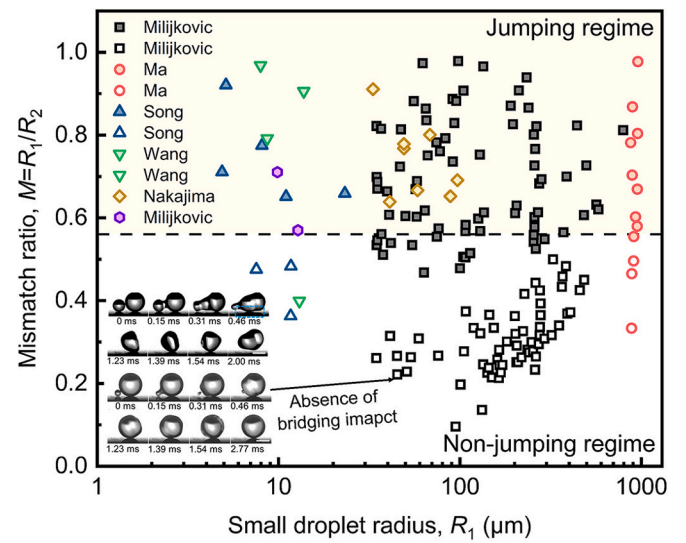


Fig. 6. The jumping state diagram of droplets affected by size mismatch is presented, with data from various studies overlaid on the graph. The legend indicates the corresponding authors of the respective studies [59,67,75,82,84,86]. Hollow markers represent no jumping, while solid markers represent jumping. The reduction in velocity originates from the asymmetric impact of the liquid bridge. When the size mismatch is significant, the absence of liquid bridge impact substrate induces jumping failure.

experiments on microstructures surfaces. Although droplet size mismatch significantly reduces energy efficiency, the combination of microscale structures and size mismatch can effectively modulate the jumping angle. Size mismatch significantly affects droplet coalescence-induced jumping at the inertial capillary scale. This leads to a particular scattering phenomenon. As the droplet size increases, the impact of size mismatch on the jump direction progressively decreases. This trend means that the coalescence of droplets with mismatched sizes cannot change the jumping direction. Only when the droplet size is close to the structural size does the droplet experience a change in the jumping direction due to the structural influence [75].

3.4. Multiple droplets coalescence

Most studies on coalescence-induced jumping focus on the coalescence of two droplets. However, in practical scenarios, droplets of varying sizes and random spatial arrangements are more common. Therefore, understanding the effects of droplet number and spatial configuration is essential for accurately modeling jumping behavior. The arrangement and number of droplets make the jumping process complex. Most of the research is still in the phase of numerical simulation. Zhang et al. [87] used molecular dynamics (MD) to examine the coalescence-induced jumping of three identical droplets under various configurations. Their results showed that a tight droplet arrangement can increase the jumping velocity by an order of magnitude. Wang and Ming [88] further demonstrated that the energy efficiency varies non-linearly with droplet arrangement. The energy efficiency fell to a minimum of about 3 % and climbed to a maximum of about 6.5 % as the distribution angle increased. Ling et al. [89] numerically investigated the effect of quantity of droplets and found that the maximum jumping height was achieved with the coalescence of four droplets. Beyond this number, gravity began to dominate, leading to a gradual reduction in jumping height. Gao et al. [90] investigated the coalescence-induced jumping of multiple nanodroplets by a molecular dynamics (MD) simulation. As the number of droplets increased, the viscous dissipation increased, and the internal flow became more complex. The energy efficiency decreased even if the droplet jumping velocity increased. To advance this field, there is a pressing need to develop novel

experimental techniques capable of reliably initiating and measuring multi-droplet coalescence-induced jumping under well-defined conditions.

3.5. Non-static coalescence

Previous experiments and simulations mostly observed the coalescence jumping in a quasi-static state, which ignores the coalescence scenario with initial velocity. The different dominant conditions of coalescence lead to differences in dynamics. Therefore, coalescence with an initial velocity significantly affects droplet jumping. The coalescence-induced jumping between a rolling droplet with initial velocity and a stationary droplet was first investigated. Due to the droplet's initial velocity breaking the coalescence symmetry, the coalescence jumping between a rolling droplet with initial velocity and a stationary droplet typically exhibits two characteristics: asymmetric developmental morphology and an inclined jumping direction [91]. When a rolling droplet coalesces at a lower initial velocity, the droplet coalescence remains governed by the inertial-capillary regime. However, the droplet rolling intensifies viscous dissipation, reducing the droplet jumping velocity. When the initial velocity of the rolling droplet is higher, inertial forces dominate over capillary forces, enhancing the jumping velocity due to increased momentum input. Li et al. [91] characterized this transition by the Reynolds number (Re). When the Reynolds number $Re < 40$, droplet coalescence follows the inertial capillary law, and the jumping velocity remains around 0.2. When the Reynolds number $Re > 40$, the jumping velocity gradually increases. Asymmetries caused by size mismatch and initial droplet motion were further analyzed in terms of energy efficiency. Interestingly, under high size mismatch conditions, the presence of an initial velocity—such as from rolling—can promote

coalescence-induced jumping by reintroducing momentum into the coalescence [92]. Moreover, off-edge rolling coalescence leads to droplet rotation. The higher the initial velocity, the greater the off-edge deviation, resulting in a lower droplet jumping velocity [93]. Gao et al. [94] found that the initial velocity of droplet motion intensified the droplet deformation. The deformation of droplets accelerated the coalescence process of the droplets and decreased the jumping velocity. Additionally, two mechanisms of jumping regulation are summarized: deformation enhancement and liquid bridge impact enhancement. The former reduces the jumping velocity, while the latter increases the jumping velocity.

4. Enhancing jumping efficiency

Despite their potential, both jumping mechanisms are inherently inefficient and uncontrollable. The droplet removal efficiency is lower than that of gravity-driven removal on flat superhydrophobic surfaces. Therefore, enhancing the jumping efficiency has become a critical challenge in applications. Currently, there are two main approaches to improving jumping efficiency: active methods, which promote droplet jumping through external field excitation, and passive methods, which effectively enhance droplet jumping efficiency through the rational design of superhydrophobic surfaces.

4.1. Active methods

When coalescence jumping occurs on superhydrophobic surfaces, the resulting trajectories tend to repel each other, indicating the presence of electrostatic interactions (Fig. 7a and b). This behavior originates from that droplets often acquire positive charges upon detachment. The

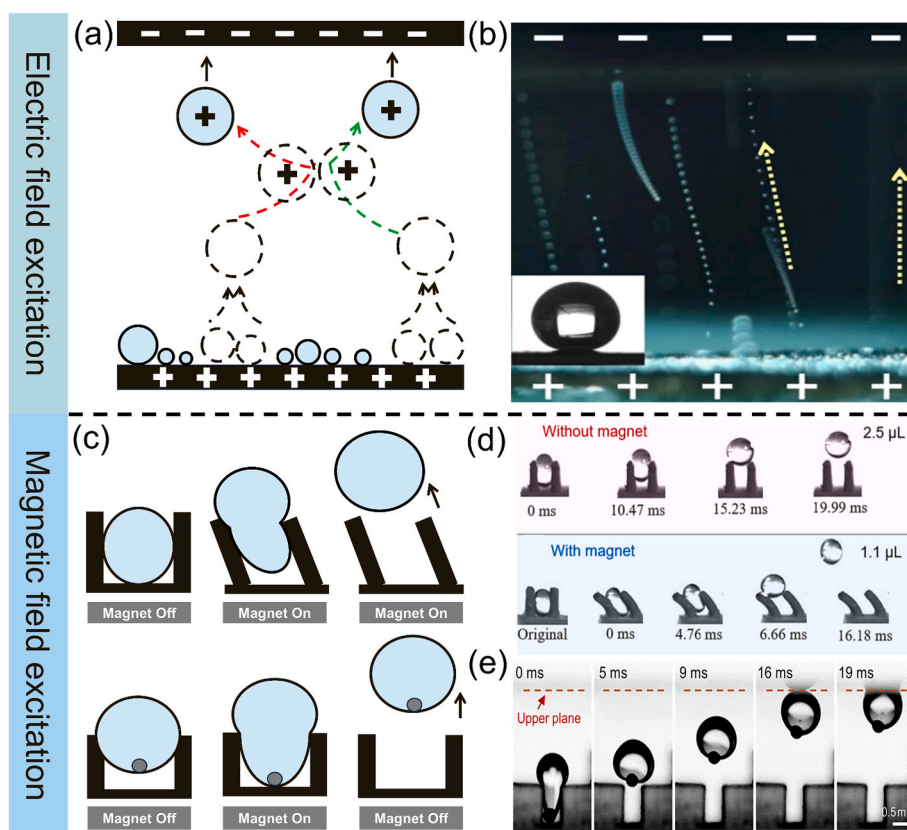


Fig. 7. External field excitation for enhancing droplet jumping. (a) schematic diagram of enhancing jumping by electric field excitation; (b) on superhydrophobic surfaces, jumping droplets experience a continuous ascent motion under the upward electric field [97]. Copyright 2024 Springer Nature; (c) schematic diagram of enhancing jumping by magnetic field excitation; Enhancing droplet jumping by magnetic excitation, which stretches the droplet to generate an upward Laplace pressure; (d) stretching magnetic droplets [101]. Copyright 2024 Elsevier; (e) compressing droplets [103]. Copyright 2024 American Chemical Society.

hydrophobic coating on the superhydrophobic surface tends to form a negative zeta potential. During the nucleation and growth of droplets, OH^- ions are adsorbed onto the coating, creating a double electric layer on the superhydrophobic surface. The rapid jumping of coalesced droplets leads to the accumulation of H^+ ions within the droplet, resulting in the jumping droplet becoming positively charged [95]. This droplet charging phenomenon has gained significant attention in recent years [96,97]. The charge on the droplet increases monotonically with the droplet's area. This means that droplets, which are more influenced by gravity, can still be driven to jump by the electric field force. For typical coalescence jumping scale, the application of a negative electric field can enhance the jumping height by approximately threefold [98]. The increased jumping efficiency is used for condensation heat transfer enhancement [99]. For cylindrical condenser tubes, applying an electric field improved the heat transfer efficiency by 50 % [100].

Another active method to promote droplet jumping is magnetic excitation. Unlike the electric field excitation mechanism, magnetic excitation indirectly alters the droplet's shape, thereby regulating excess surface energy to promote droplet jumping (Fig. 7c). Zhang et al. [101] fabricated flexible magnetic micropillars that deform under an external magnetic field to compress droplets, thereby reducing their surface area. This geometric confinement results in the release of excess surface energy, which is subsequently converted into kinetic energy to initiate jumping (Fig. 7d). With the application of the external magnetic field, the detachment radius of the droplet is reduced to half of its original size, allowing for efficient removal of droplets with radius more significant than $10\text{ }\mu\text{m}$ [102]. Huang et al. [103] placed ferromagnetic particles into the droplet. The application of a magnetic field manipulated the morphology of droplet through particle stretching, thereby modulating surface energy to induce jumping (Fig. 7e). This method achieved single-droplet jumping with an energy efficiency of 0.33, representing an approximately 400 % improvement. In addition to enhancing energy conversion, magnetic excitation also enables precise control of the jumping direction in three dimensions.

4.2. Passive methods

The passive method primarily improves the efficiency of droplet jumping by designing and fabricating superhydrophobic surfaces with intricate structures. As this approach does not rely on external field excitation, it is referred to as passive in contrast to active methods.

4.2.1. Macro structure surface

Macrostructures are large-scale, visually discernible geometric features typically with millimeter dimension. In studies of macrostructures, droplets are usually introduced onto the surface by deposition methods allowing researchers to observe and analyze their dynamic behavior conveniently. Recent research has demonstrated that surfaces with macrostructures can effectively regulate the direction of droplet motion and enhance jumping velocity (Fig. 8 and Table 2). Droplet jumping converts excess surface energy into kinetic energy through energy dissipation. Therefore, jumping efficiency can be improved through two main strategies. One method involves reducing the dissipation generated during the jumping process. The alternative is raising the excess surface energy produced before jumping.

A triangular prism macrostructure surface first achieved a maximum dimensionless droplet jumping velocity of 0.53 and an energy efficiency of 22 % [104]. The jumping velocity has been enhanced by approximately threefold. The mechanism enhancing the jumping is that the conical structure accelerates the contraction of the critical region at the droplet-substrate interface, increasing the jumping velocity of the droplet. Numerical studies have investigated the effects of related geometric parameters, including prism shape, apex angle, and height [105]. Results showed that smaller apex angles and greater height lead to higher jumping velocity. Vahabi et al. [106] fabricated superhydrophobic conical surfaces with a smaller apex angle, achieving a

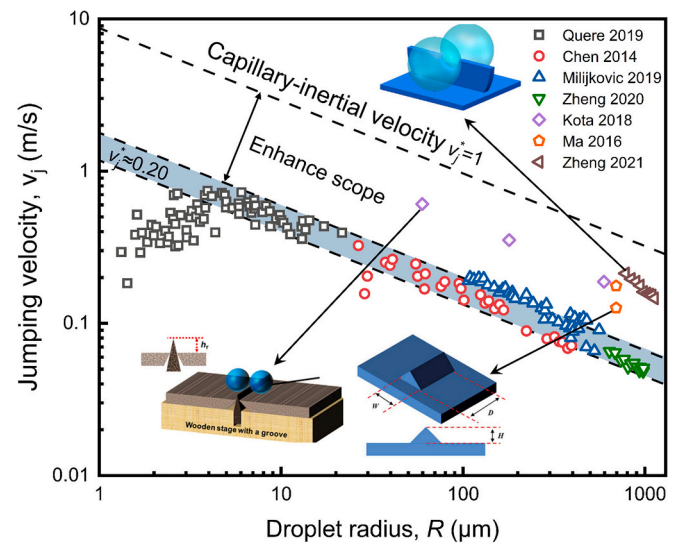


Fig. 8. Measured values of the velocity of jumping droplets as a function of the coalescence droplet radius for same-sized droplet pairs and composite results for enhanced jumping by coalescence on macrostructures. The legend indicates the corresponding authors of the respective studies [72,73,76,77,101,103,104].

Table 2
Effect of macrostructure for jumping performance.

structures	Dimensionless jumping velocity v_j^*	Energy efficiency η	Jumping direction θ
Flat [75]	0.20	6 %	90°
V- Groove [81]	0.61	30 %	30° - 60°
Triangular prism [104]	0.53	22 %	90°
Ridge [106]	0.60	19 %	90°
U- Groove [107]	0.70	43 %	17° - 90°
Egg tray [110]	0.78	52 %	90°
Circular groove [111]	0.63	44 %	60° - 90°
Grove [43]	0.74	46 %	90°
Rectangular slot [112]	0.74	48 %	60° - 90°
Wall [117]	0.35	18 %	45° - 90°
Cylinder [118]	0.56	32 %	30° - 90°

dimensionless jumping velocity of 0.70 and energy efficiency of 33 %, attributing the improvement to liquid bridge reshaping and velocity vector redistribution. In addition to the conical structure, U-shaped groove structures [107], curved surface structures [108], and fiber [109] have also been studied. These structures are designed to guide the flow field and adjust momentum redistribution, further enhancing jumping performance. Beyond redirecting velocity vectors, another strategy focuses on increasing excess surface energy prior to jumping. The basic method for increasing surface energy is to manufacture inward-curved arc-shaped structures that enlarge the solid-liquid contact area. Yuan et al. [110] utilized machine learning to create an ideal egg tray macrostructure to increase excess surface energy. The optimal dimensionless jumping velocity with an egg tray structure reached 0.78, and the energy efficiency reached 52 %. Liu et al. [111] employed an annular groove structure to increase liquid bridge area while suppressing droplet vibration, thus reducing viscous dissipation. Some researchers have developed structural surfaces that efficiently jump by combining droplet coalescence and Laplace pressure difference. A grooved surface is employed to utilize the effects of Laplace pressure difference and velocity vector redistribution, resulting in a dimensionless jumping velocity of 0.74 and an energy efficiency of 46 % [43]. Liu et al. [112] designed a superhydrophobic structure that achieved a velocity of 0.78 and efficiency of 48 % by manipulating Laplace pressure

gradients.

In addition to enhancing jumping velocity, another efficient droplet detachment method is to control the directional jumping of droplets and sweep tiny droplets along the path of the jumping droplets [113–116]. Currently, asymmetric structures are the mainstream method for achieving directional droplet jumping. V-groove [81] first achieved a 60° vertical inclination angle with a dimensionless jumping velocity of 0.61. Simulations revealed that sidewalls restricted lateral expansion, reducing viscous dissipation and enhancing vertical momentum transfer. Furthermore, V-grooves shortened the liquid bridge impact time, improving energy transfer efficiency. Yuan et al. [117] managed the droplet's directional jumping by limiting its motion path and examining how the Bo number affected the jumping direction, revealing that controlling the interface between the liquid bridge and the surface is necessary to improve the efficiency of droplet jumping. Gao et al. [118] constructed an inclined cylindrical macrostructure with a vertical inclination angle of 68° and an energy efficiency of 32 % to achieve

droplet jumping.

Although progress has been made, achieving both high-speed and directional jumping remains a challenge. The mechanism for improving droplet jumping efficiency mainly focuses on reducing the coalescence time to reorient the upward velocity vector. Viscous or oscillatory dissipation has not been eliminated, leaving a significant gap to reach the theoretical maximum velocity. Additionally, directional jumping is achieved through asymmetric structures, which cause asymmetric coalescence. Direction control of droplet jumping is mainly confined to within 90 degrees; achieving bidirectional jumping remains a significant challenge.

4.2.2. Array structure surface

Macroscopic structures typically fall within the range discernible to the naked eye, with droplet radius generally ranging from $500\ \mu\text{m}$ to $1000\ \mu\text{m}$ (Figs. 8 and 9a). These droplet sizes are considered relatively large and have limited practical application. Therefore, macroscopic

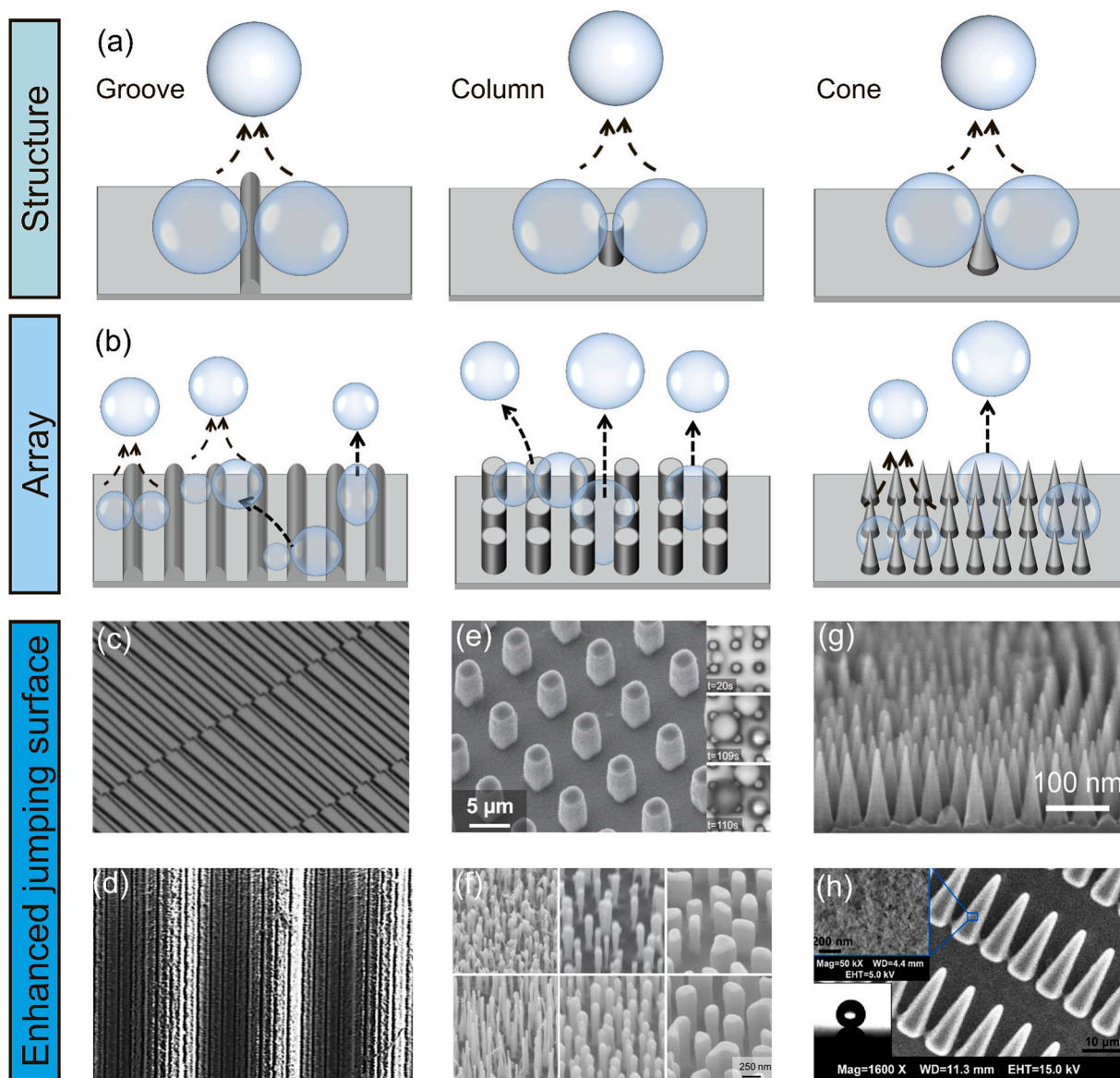


Fig. 9. Surfaces with micro-array structures enhanced droplet jumping. (a) schematic diagram of jumping enhancing by macrostructure; (b) schematic diagram of jumping enhancing by micro-array structure; (c) image of the asymmetric ridge array [124]. Copyright 2025 Elsevier; (d) SEM image of two roughness ridge array [126]. Copyright 2024 American Chemical Society; (e) enhancing tiny droplet jumping by microcolumn [127]. Copyright 2015 American Chemical Society; (f) SEM images of the microcolumn array for dewetting. [128] Copyright 2017 American Chemical Society; (g) SEM image of conical array. [80] Copyright 2019 American Physical Society; (h) annularly arranged conical arrays for directional droplet transport [130]. Copyright 2023 American Chemical Society.

structure research primarily aims to clarify the mechanisms for enhancing jumping. To bridge the gap between experimental demonstration and application, one common approach is to scale down macroscopic structures to match the typical size of jumping droplets and fabricate regular arrays. At the micro/nanoscale, surface geometry becomes a decisive factor in determining wettability characteristics [119,120]. Surfaces with specific microstructured features often exhibit high apparent contact angles, favoring droplet mobility and jumping. Structures such as pillars, ridges, and cones have been extensively demonstrated to improve jumping performance. Protruding structures facilitate the early coalescence of droplets and momentum redirection. Furthermore, the curvature difference of these structures induces a Laplace pressure difference, which promotes droplet dewetting and even jumping. The combined effects of these two mechanisms contribute to the enhanced jumping functionality of the macrostructures above (Fig. 9b).

Inspired by macrostructures, cone, groove, and pillar arrays were proposed as advantageous to droplet jumping [121–123]. The groove structure is widely used to enhance droplet jumping performance due to its easy manufacturability. The underlying mechanism for the application is the premature collision of the liquid bridge with the substrate. The droplet jumping probabilities for flat surfaces, groove arrays, and staggered groove arrays were 30 %, 40 %, and 50 % (Fig. 9c) [124], respectively. Ma et al. [125] achieved 100 % jumping probability for droplets larger than 10 μm by designing a grid-like groove surface. Wan et al. [126] further reduced the detachment size by 50 % using a dual-layer groove structure fabricated by laser etching (Fig. 9d).

Unlike groove array surfaces, microcolumn (micropillar) array surfaces primarily influence jumping by modulating wetting states. The confinement effect of micropillars facilitates a transition from the Wenzel state to the Cassie state, reducing droplet adhesion and enhancing jumping efficiency (Fig. 9e) [127]. Zhang et al. [87] verified that the droplets trigger a dewetting transition from the Wenzel state to the Cassie state when the distance between the micropillar arrays changes. Droplet growth process in micropillar arrays follows a three-stage model which shows that achieving a Cassie state requires a nucleation density lower than the structural density to avoid flooding. Therefore, the critical droplet size for jumping is largely determined by the geometric scale of the micropillars (Fig. 9f). Mulore et al. [128] validated the accuracy of the theoretical model and established a function of the critical jumping size and surface topology. Furthermore, Laplace pressure gradients induced by geometric constraints in micropillar arrays further assist in the passive wetting state transition and jumping. Numerous studies have examined the influence of micropillar height and spacing on jumping performance. Numerical results suggest that width and spacing are the dominant factors, while height has a less significant effect [129]. The effect of the microarray structure surface on the droplet jump varies with changes in the geometric properties of the array structure and droplet. When handling multi-parameter optimization problems, artificial intelligence can be used to predict and optimize the geometric structure of the array to maximize the effect of droplet jumps.

Lecointre et al. [80] achieved 1 μm droplets jumping by conical microstructure arrays, effectively reducing the critical size to one-tenth of its original value (Fig. 9g). To further regulate jumping direction, Hou et al. [130] fabricated a conical surface with an annular distribution (Fig. 9h). The curvature difference induced by the micro-cones led to asymmetric pressure distribution, causing droplets to migrate along the gradient direction. Due to the fabrication challenges of conical arrays, it is more feasible to use numerical simulation to study the impact of the geometric properties of the cone array on the droplet jumping. Shi et al. [131,132] varied the cone array's height and spacing to optimize the structural surface by numerical simulation. Results showed that the droplet jumping height rises as cone height increase and cone-to-cone distance decrease. The droplet jump height is determined mainly by the cone height. Droplet jumping can be improved by decreasing the

cone top angle. The improved cone array allows for a 245 % increase in jumping velocity and jumping of low contact angle (100°) [133].

4.2.3. Hierarchical structure surface

Compared to single-roughness macrostructures or arrayed structures, surfaces with hierarchical structures exhibit a better jumping enhancement effect. Currently, hierarchical surfaces are mainly categorized into two types. One type is a structure with dual roughness, called structural hierarchical surfaces [134–136]. The other type is the biphilic surface, formed by alternating hydrophilic and hydrophobic patterns (Fig. 10a) [137–139].

According to the classical nucleation theory, droplet nucleation predominantly occurs at locations with lower energy barriers, such as hydrophilic surfaces [138,140]. This characteristic significantly limits the nucleation rate on superhydrophobic surfaces, thereby constraining the practical application of droplet jumping. In contrast, biphilic surfaces which consist of alternating hydrophilic and hydrophobic regions, provide a promising solution by integrating the advantages of both wetting states. For biphilic surfaces, the hydrophobic regions reduce droplet pinning, while the hydrophilic regions tend to facilitate faster droplet nucleation (Fig. 10b) [140]. Adjusting the intrinsic surface wettability or the liquid surface tension can influence the nucleation and mobility of droplets on the surface. [141] The nucleation enhancing of biphilic surfaces have been experimental demonstrated. The total droplet volume accumulated on biphilic surfaces can exceed that on homogeneous hydrophobic surfaces by more than twofold (Fig. 10c) [142]. However, nucleation randomness on heterogeneous surfaces presents a major challenge, making precise spatial control of droplet formation difficult to achieve. The adhesion force caused by hydrophilic spots significantly reduces the jumping velocity on biphilic surfaces. [143] Finding a balance between the high adhesion caused by hydrophilic surfaces and the increased droplet nucleation is another major challenge for applying biphilic surfaces in droplet jumping.

Hierarchical structured surfaces typically feature dual roughness levels. The nanometer-scale roughness corresponds to the size of nucleated droplets and facilitates the Wenzel-to-Cassie wetting transition by promoting droplet dewetting through nanostructures. In contrast, the micrometer-scale roughness aligns with the typical scale of droplet jumping, enhancing jumping dynamics through structured arrays [144–146]. The synergy between the dual roughness levels contributes to the overall enhancement of droplet jumping. Hierarchical surfaces with mesh-like [147,148], columnar [149,150], and conical [134,136] configurations have been extensively studied for their ability to improve jumping efficiency. The large solid-liquid contact area in mesh structures (Fig. 10d) effectively promotes droplet growth within the pores. The confinement effect induces a top-heavy morphology, allowing for efficient detachment by single-droplet jumping and coalescence jumping [147]. In addition to mesh structure surface, hierarchical ridge structures achieve rapid droplet growth and efficient detachment by coupling small-scale droplet shedding and transport with large-scale droplet detachment (Fig. 10e). The growth rate of microdroplets increases by 20 %, while the droplet size distribution narrows by 30 % [151]. However, the difficulty of nucleation imposes significant limitations on droplet jumping. The imbalance between droplet supply and demand during the jumping process dramatically restricts the development of hierarchical structured surfaces.

Recent studies have shown that surfaces coupling both dual-scale roughness and biphilic characteristics significantly enhance droplet jumping performance [138,152]. The alternating hydrophilic and hydrophobic patterned arrangement promotes droplet nucleation, ensuring droplet growth rates, while the hierarchical structure facilitates droplet shedding and jumping across multiple scales. Combining these two mechanisms further improves the efficiency of droplet jumping. Droplets preferentially nucleate on hydrophilic regions, and the Laplace pressure generated by the droplets stretching through inclined conical structures drives droplet movement and shedding

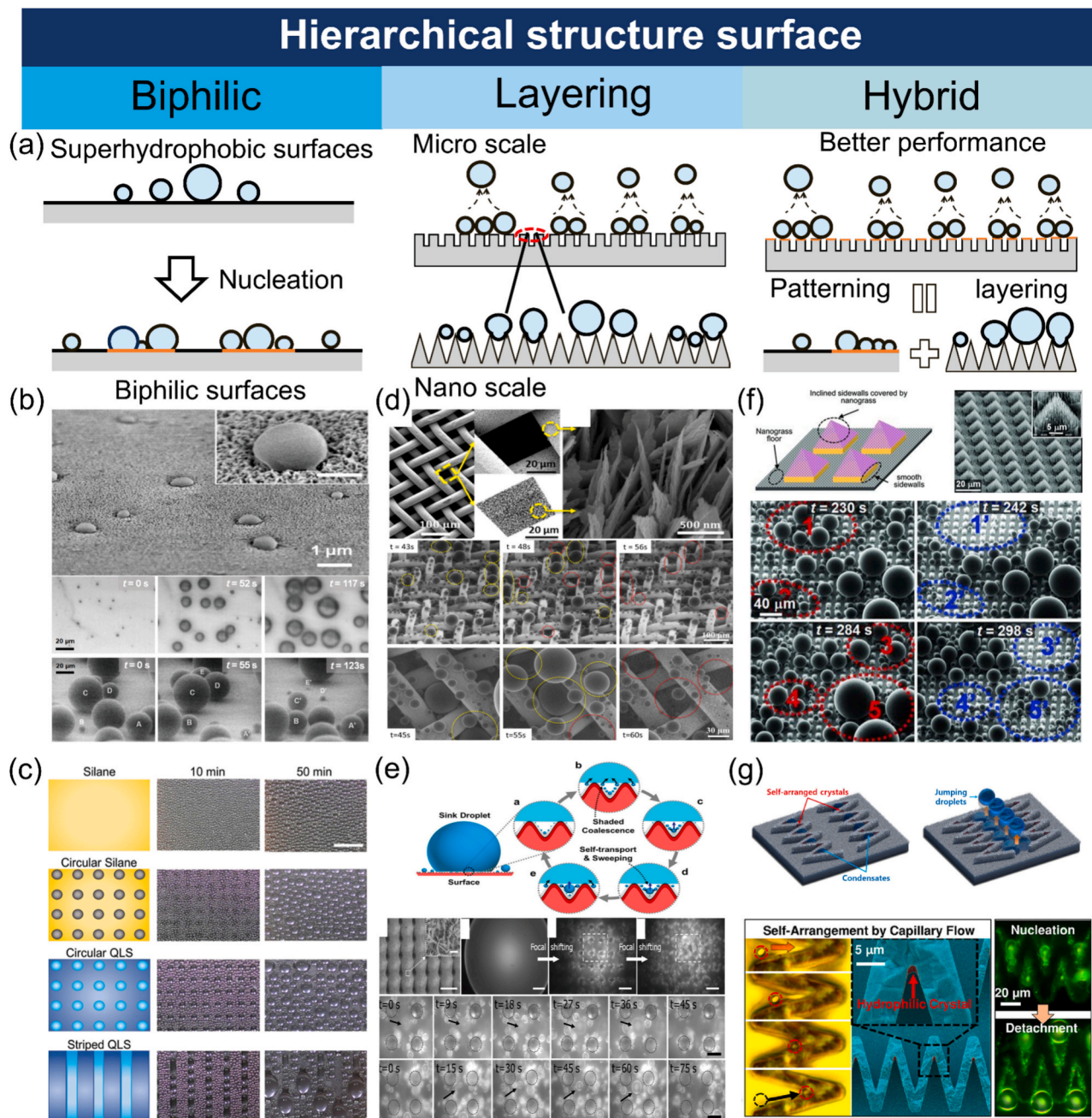


Fig. 10. Surfaces with hierarchical structures enhanced droplet jumping. (a) surface modification to enhance jumping: basic approaches include wettability alteration and multi-scale structures; (b) biphilic surfaces accelerate droplet nucleation [140]. Copyright 2018 American Chemical Society; (c) patterned design achieves high water collection efficiency [142]. Copyright 2023 John Wiley and Sons; (d) mesh-like multilayer structures drive efficient droplet jumping [147]. Copyright 2016 American Chemical Society; (e) hierarchical condensation with droplets detaching via gravity [151]. Copyright 2020 American Chemical Society; (f) biphilic surfaces combined with hierarchical structures enable efficient droplet nucleation and jumping [139]. Copyright 2011 John Wiley and Sons; (g) z-shaped structures drive directional droplet movement and jumping, with hydrophilic spots positioned within the Z-shape [153]. Copyright 2021 Elsevier.

(Fig. 10f). The multidimensional span of droplet jumping and movement enhances the diversity of structural design. Both unidirectional [139] and opposite-direction [153] jumping and movement achieve effective droplet detachment (Fig. 10g). Additionally, due to heterogeneous structures favoring nucleation at hydrophilic sites, the fragile hydrophobic surface is protected, extending its service life and enhancing its robustness.

4.2.4. Bioinspired surface

Droplets frequently display different wettability, shapes, and dynamic behaviors on biological surfaces, inspiring the design of

biomimetic hierarchical structures. Lotus leaf known for its intrinsic hydrophobicity has been widely studied since the early stages of research on superhydrophobic surfaces [74]. The first observation of coalescence-induced droplet jumping on a biological surface was reported on the lotus leaf's dual-roughness structure (Fig. 11a). This observation prompted detailed investigations into viscous dissipation and critical size, driving the advancement of droplet jumping mechanisms. Although the hydrophobic structure of the lotus leaf itself has a limited effect on promoting droplet jumping, it laid the foundation for subsequent studies on the continuous dynamic of droplet jumping. Subsequent studies extended to various plants and animals. By

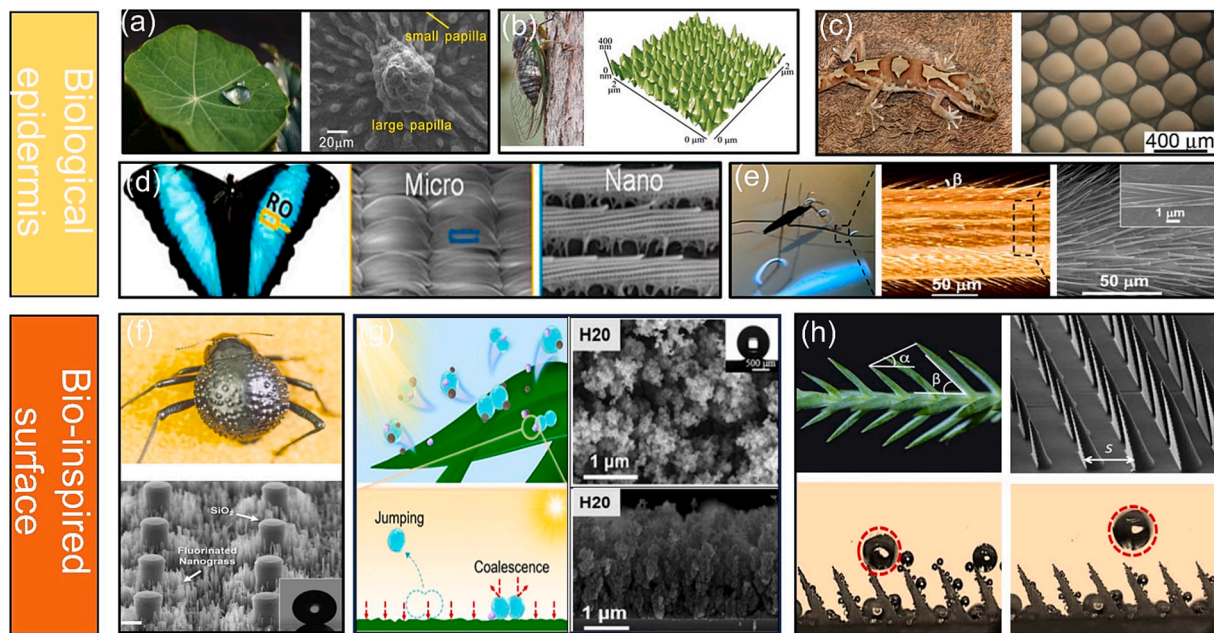


Fig. 11. Bionic Surfaces enhanced droplet jumping. (a) double roughness lotus leaf [74]. Copyright 2013 AIP Publishing; (b) superhydrophobic cicada wings with self-cleaning functions [8]. Copyright 2013 National Academy of Sciences; (c) geckos and its hierarchical nanostructures [154]. Copyright 2015 Elsevier; (d) butterfly wings with the function of directional droplet transport [156]. Copyright 2014 American Chemical Society; (e) water strider and its microscale structure [157]. Copyright 2015 National Academy of Sciences; (f) multi-hierarchical nano grass structures inspired by beetle [138]. Copyright 2015 American Chemical Society; (g) solar-powered frost-resistant surfaces inspired by wheat [159]. Copyright 2021 National Academy of Sciences; (h) asymmetric multilayer surface structures inspired by pine needle [160]. Copyright 2020 AAAS.

observing the phenomenon of droplet jumping on cicada wings removing contaminants (Fig. 11b), droplet jumping on cicada wings can be categorized into three types: floating, lifting, and coalescing [8]. Although the droplet jumping phenomena differ, the fundamental principle remains the same—excess surface energy generated during droplet coalescence provides the energy for jumping. The feet of geckos have also garnered attention for their low adhesion and self-cleaning properties, which are attributed to micrometer-scale nipple-like structures [154,155]. These structures stretch droplets and create Laplace pressure differences, enabling efficient detachment (Fig. 11c). Similarly, butterfly wings exhibit self-cleaning phenomena, where coalesced droplets also achieve self-cleaning effects on static wings [156]. Experimental results reveal that fog droplets grow on the asymmetric micro-ratchet structures of butterfly wings, generating unbalanced surface tension that drives directional movement and jumping of the droplets (Fig. 11d). Water striders demonstrate another fascinating case, where droplets grow along the microhair gradient on their legs and detach due to Laplace pressure-driven jumping (Fig. 11e) [157]. Even in pathological contexts, such as rust-infected plant leaves, coalescence jumping of droplets carrying fungal spores plays a key role in aerosol-assisted disease transmission [158]. The aforementioned biological surfaces typically possess dense structures at the micron scale, which account for their low adhesion properties. These microstructures often resemble macroscopic structures that promote droplet jumping, thereby facilitating the jumping behavior of droplets.

In addition to the droplet jumping directly observed on plant and animal surfaces, researchers have also drawn inspiration from nature to design various biomimetic structured surfaces that enhance droplet jumping. Inspired by the water-harvesting phenomenon of Namib beetles, which utilize asymmetrical hydrophilic and hydrophobic patterns on their backs, a hierarchical structure was fabricated (Fig. 11f) [138]. The structure places hydrophilic patches atop micropillars surrounded by superhydrophobic nanoglass, achieving high-density droplets jumping below 10 μm . Droplets even jump on extremely hydrophilic regions. Inspired by the non-wetting properties of wheat leaves,

bioinspired superhydrophobic surfaces were fabricated using laser deposition (Fig. 11g) [159]. By constructing specialized nano-hierarchical structures, bioinspired surfaces reduce the solid-liquid contact area and adhesion between droplets and the surface, thereby enhancing droplet jumping efficiency. Compared to the copper surface, the water removal rate on the bioinspired surface increased by 800 %. On Janus structured surfaces inspired by pine needles [160] (Fig. 11h), the asymmetric tip-induced flipping (TIF) effect resulted in spontaneous, directional droplet ejection.

5. Applications

The optimization of droplet jumping holds significant practical application value. Strategies such as improving jumping velocity, controlling jumping direction, and reducing the critical jumping radius aim to enable practical applications of droplet jumping. Enhanced condensation heat transfer is a fundamental application of droplet jumping, allowing efficient heat transfer by achieving the low thermal resistance of dropwise condensation. Self-cleaning applications can be achieved using droplet jumping to remove contaminants from surfaces. Stimulating the jumping and detachment of condensation droplets before freezing enables anti-icing and anti-frosting functionalities.

5.1. Enhancing condensation heat transfer

Dropwise condensation exhibits significantly higher heat transfer efficiency than filmwise condensation. [161] The high heat resistance is caused by a thick liquid film, which is the outcome of film condensation. In dropwise condensation, droplets must be periodically removed from the surface once they reach a critical size to keep the condensation process going stably and avoid flooding [162]. On vertical surfaces, gravity is typically relied to induce droplet detachment. A direct proportionality exists between the capillary length, l_c , and the critical detachment diameter, D_c . The approximate value of D_c for water is 2.7 mm. For steam condensation at a vapor pressure of 100 kPa, the heat

transfer coefficient (h) is typically about $100 \text{ kW/m}^2\cdot\text{K}$ [163]. According to theoretical models, the heat transfer coefficient rises significantly as D_c decreases. The model can be validated by reducing detachment size by a factor of ten ($D_c \sim 0.1 \text{ mm}$) and increasing the condensation heat transfer at $P_v \approx 100 \text{ kPa}$ to about $h \approx 500 \text{ kW/m}^2\cdot\text{K}$ by exposing the condenser to external forces such as steam shear or centrifugation. Therefore, reducing D_c is essential to improving heat transfer from condensation. [164] However, achieving D_c below the capillary length typically requires external stimulation, which introduces additional complexity and limits applications.

The phenomena of droplet coalescence-induced jumping during dropwise condensation on superhydrophobic surfaces was first observed by Borekyo and Chen [16]. Compared to gravity-driven droplet departure, coalescence jumping reduces the critical detachment D_c by approximately one-third, offering exciting potential for enhancing condensation heat transfer. However, Subsequent experiments have shown that dropwise condensation in the droplet jumping mode has a relatively lower heat transfer coefficient than film condensation [165]. A significant breakthrough was made by Miljkovic et al. [6] They used copper oxide to create knife-shaped nanostructured tubes and put the tube through a vacuum steam chamber (Fig. 12a). According to the results, droplet jumping condensation has a total heat flux that is 25 % higher and a heat transfer coefficient that is 30 % higher than conventional dropwise condensation, with heat transfer coefficients of about $100 \text{ kW/m}^2\cdot\text{K}$ and $70 \text{ kW/m}^2\cdot\text{K}$, respectively, when the supersaturation

conditions are low, especially at a subcooling degree of less than 5°C and corresponding vapor pressure of 3 kPa . However, at high subcooling, the nanostructures flood with water, which results in the formation of secure pinning Wenzel droplets. As a result, the condensation heat transfer coefficient is significantly reduced (40 %) compared to conventional dropwise condensers. Based on the studies above, hierarchical structured finned tubes have been fabricated to enhance condensation efficiency [166]. At the microscale, porous structures optimize droplet nucleation and shedding, at the macroscale, fins improve droplet jumping efficiency and increase the heat transfer area (Fig. 12b). Nevertheless, the enhancement effect diminishes at high subcooling due to persistent flooding. Furthermore, OH^- ions are adsorbed on the superhydrophobic surface during the nucleation and development of droplets, increasing the net H^+ ions inside the droplets. The condensed droplets that developed on the superhydrophobic surface had a positive charge at the macroscopic level. An outside electric field repels the droplets as a result, encouraging the droplets to detach (Fig. 12c). [95] By actively applying negative charges to a cylindrical copper mesh surrounding the condensation tube, the heat transfer coefficient of dropwise condensation can be increased by at least 50 % [100,167]. However, the failure of dropwise condensation under high subcooling conditions remains a significant factor limiting its application.

At low subcooling, the advantages of droplet jumping condensation have been well demonstrated. However, the failure of superhydrophobicity due to flooding has remained a significant limit for

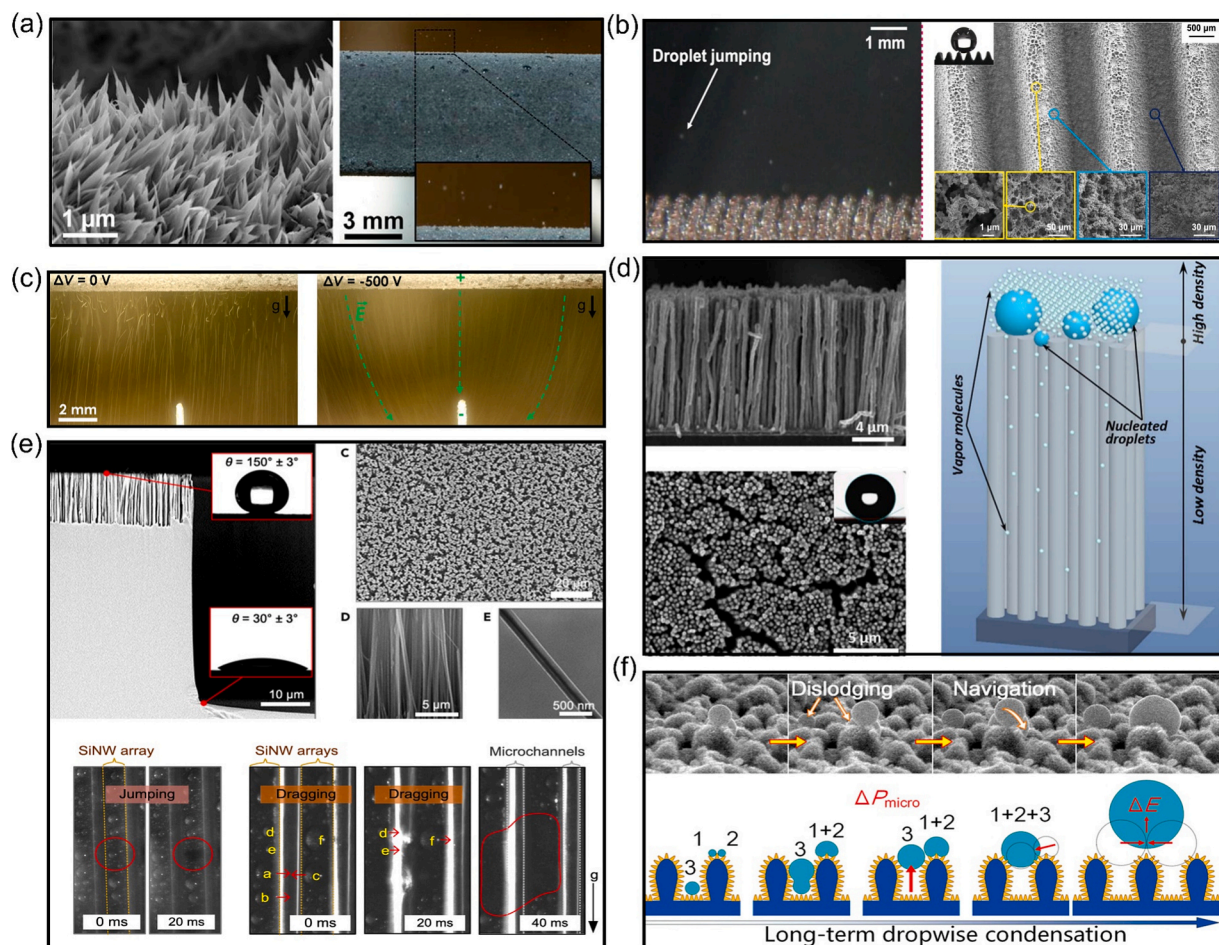


Fig. 12. The applications of droplet jumping in heat transfer. (a) droplet jumping condensation on knife-shaped nanostructured SHS [6]. Copyright 2013 American Chemical Society; (b) multiscale modified surfaces for enhancing condensation heat transfer [166]. Copyright 2023 John Wiley and Sons; (c) electric field excitation for enhancing condensation jumping [95]. Copyright 2013 Springer Nature; (d) nanowire networks enable stable and efficient heat transfer under high subcooling conditions [168]. Copyright 2017 Elsevier; (e) hierarchical structures and nanowires enhanced long-term droplet jumping condensation [169]. Copyright 2019 Elsevier; (f) multilayer structural modifications enable robust dropwise condensation [170]. Copyright 2021 American Chemical Society.

jumping droplet condensation. Flooding typically occurs under conditions of high subcooling or long-term condensation. As the amount of condensate increases, the volume of droplets removed by jumping becomes lower than that generated by nucleation, leading to the coverage of superhydrophobic micro/nanostructures by condensate. To address this challenge, researchers have explored approaches for achieving stable jumping-mode condensation. One of the most promising strategies involves the application of densely packed nanowire arrays, which limit vapor infiltration into the surface and promote preferential nucleation at the nanowire tops. Due to the tight spacing between nanowires, supersaturated vapor cannot easily penetrate into the interstitial gaps, thereby maintaining stable condensation (Fig. 12d). Nanowire arrays enable stable droplet jumping even under significant vapor pressures $P_v \approx 60$ kPa and substantial subcooling $\Delta T > 25$ °C [168]. Wen et al. [5] further advanced this concept by fabricating nanowire arrays with varied heights, enabling Laplace pressure-induced dewetting and preferential nucleation on shorter structure. Stepped configuration delayed flooding and enhanced the condensation heat transfer coefficient. Further studies have shown that when using three-dimensional superhydrophobic nanowires, the subcooling of steady condensation can reach 28 K. By interconnecting nanowire structures, the nucleation sites can be spatially controlled, resulting in a 100 % increase in heat flux compared to the most advanced hydrophobic surfaces. Additionally, hierarchical grooved structures with nanowire arrays significantly enhance condensation heat transfer under high subcooling conditions by combining dropwise and filmwise condensation (Fig. 12e) [169]. The surface achieves high heat transfer efficiency at low subcooling levels through droplet jumping driven by nanowires at the groove tops and liquid film sliding within the grooves. Under high subcooling conditions, droplet coalescence induced by nanowires replenishes the liquid film within the grooves, improving surface performance and preventing flooding.

When dropwise condensation on superhydrophobic surfaces leads to rapid phase change heat transfer and droplet detachment, this process is associated with several undesirable features. The energy barrier for nucleation is superior on hydrophobic surfaces compared to hydrophilic surfaces, negatively impacting nucleation density. Furthermore, due to interfacial adhesion and pinning effects, droplets on superhydrophobic surfaces often fail to jump. To address these challenges, researchers developed biphilic surface that promotes dropwise condensation. In biphilic surfaces, hydrophilic domains enhance nucleation density, while superhydrophobic domains facilitate droplet jumping, achieving a balance between droplet formation and removal. Chen et al. [139] fabricated the biphilic pyramid surface that promotes dropwise condensation and dropwise jumping. The surface is covered by nanoglass micro-pyramids. There is no nanoglass present in the sidewalls portion of the micro-pyramid. Initially, droplets usually nucleation on the sidewalls at the micro-pyramid base. Then, as they get bigger, they hang and jump at the top of the cone. Recently, a heat transfer coefficient of $h \approx 218$ kW/m²·K ($\Delta T = 0.2$ °C) was achieved using hierarchical superhydrophobic surfaces composed of micro-pyramidal structures (Fig. 12f) [170]. The microstructures facilitated droplet jumping driven by Laplace pressure differences between the pyramids, enabling continuous condensation droplet jumping. However, this record remains significantly below the fundamental heat transfer limit for coalescence-induced jumping droplets. Under conditions of high subcooling, flooding caused by rapid nucleation in hydrophilic regions remains a challenge in the design of biphilic condensation surfaces. Designing surface structures that enhance condensation while minimizing flooding and thermal resistance remains a crucial research direction for enhancing dropwise condensation.

5.2. Self-cleaning

After the solid surface is coated with superhydrophobicity, water vapor condenses to form droplets and convey dust particles. The

droplets detach from the solid surface by jumping to perform the self-cleaning function. The self-cleaning surfaces were found in natural species, including cicadas, geckos, rice, and lotus leaves. Some researchers have created bionic self-cleaning surfaces inspired by the abovementioned creatures. Recent studies have demonstrated self-cleaning capabilities on surfaces modeled by clover leaves and butterfly wings, achieving particle removal by droplet–particle coalescence jumping (Fig. 13a) [171,172]. Notably, this mechanism differs from conventional droplet–droplet coalescence jumping. Droplet–particle coalescence enables the removal of particles in the 10–100 μ m range through direct interaction between droplets and solid contaminants. Additionally, inspired by the conical micro-nanostructures of compound eyes and insect wings, researchers have fabricated highly transparent biomimetic self-cleaning surfaces (Fig. 13b) [173]. On bionic compound eyes surface, pollutants are rapidly removed by coalescence jumping (Fig. 13c).

In addition to biomimetic self-cleaning surfaces, superhydrophobic transparent glass for self-cleaning solar photovoltaic panels has been fabricated through surface modification of borosilicate glass. For macroscopic contaminants, self-cleaning is primarily achieved by wind or gravity. For tiny dust particles, coalescence jumping of condensed droplets provide effective removal (Fig. 13d) [174]. To achieve a self-cleaning feature, Geyer et al. [9] deposited siloxane nanowires on glass slides and applied surface fluorination to reduce surface energy. Outdoor tests on coated polyester fabrics revealed that condensation droplets could autonomously clean the surface without external intervention. However, the self-cleaning ability of droplet jumping is generally confined to particles larger than the nanogap spacing but smaller than capillary-scale. Microscopic observations have shown that particles significantly larger than the nanowire gap width are effectively removed from superhydrophobic nanowire by droplet jumping. However, the superhydrophobic nanowire surface retained most particles with dimensions like the nano-gap width (Fig. 13e) [175].

The research and investigation of superhydrophobic materials and structures has aided in quickly developing surfaces that can clean themselves. These days, self-cleaning coatings are widely used in vital applications such as solar panels, glass curtain walls, automobile surfaces, and textiles. The dropwise jumping self-cleaning method's use is still constrained by the modest removal force of individual droplets since capillary forces play a significant role in droplet jumping at the micrometer scale.

5.3. Anti-icing and anti-frosting

In cold and humid environments, water vapor preferentially condenses into liquid droplets or films rather than forming frost directly, since the energy barrier for nucleating supercooled liquid embryos is slightly lower than that for ice embryo formation. If the condensed liquid is not removed in time, it can eventually freeze, leading to frost or ice accumulation on the solid surface. Smaller droplets on the solid surface remain adhere, while larger droplets detach under the influence of gravity, airflow, and vibration. The jumping of microscopic droplets surface are vital in preventing frost formation. The researchers modified the superhydrophobic structure to remove condensed droplets, encouraged droplet jumping, and achieved anti-frost and anti-icing properties.

Micro-cone and micro-needle arrays, which resemble the surface that increases the droplet jumping height, are commonly used in deicing applications. On superhydrophobic micro-tower array surface, condensed droplets migrated upward independently, and the Wenzel-Cassie state transition was realized [176]. The condensed droplets migrate upward spontaneously through Laplace pressure, and the tower array's pointed top lowers the adhesion of condensed droplets, enabling quick surface deicing (Fig. 14a). The photothermal surface featuring micro-papillae, inspired by wheat leaves, exhibits anti-icing functionality [159]. Even at temperatures as low as -50 °C, the surface

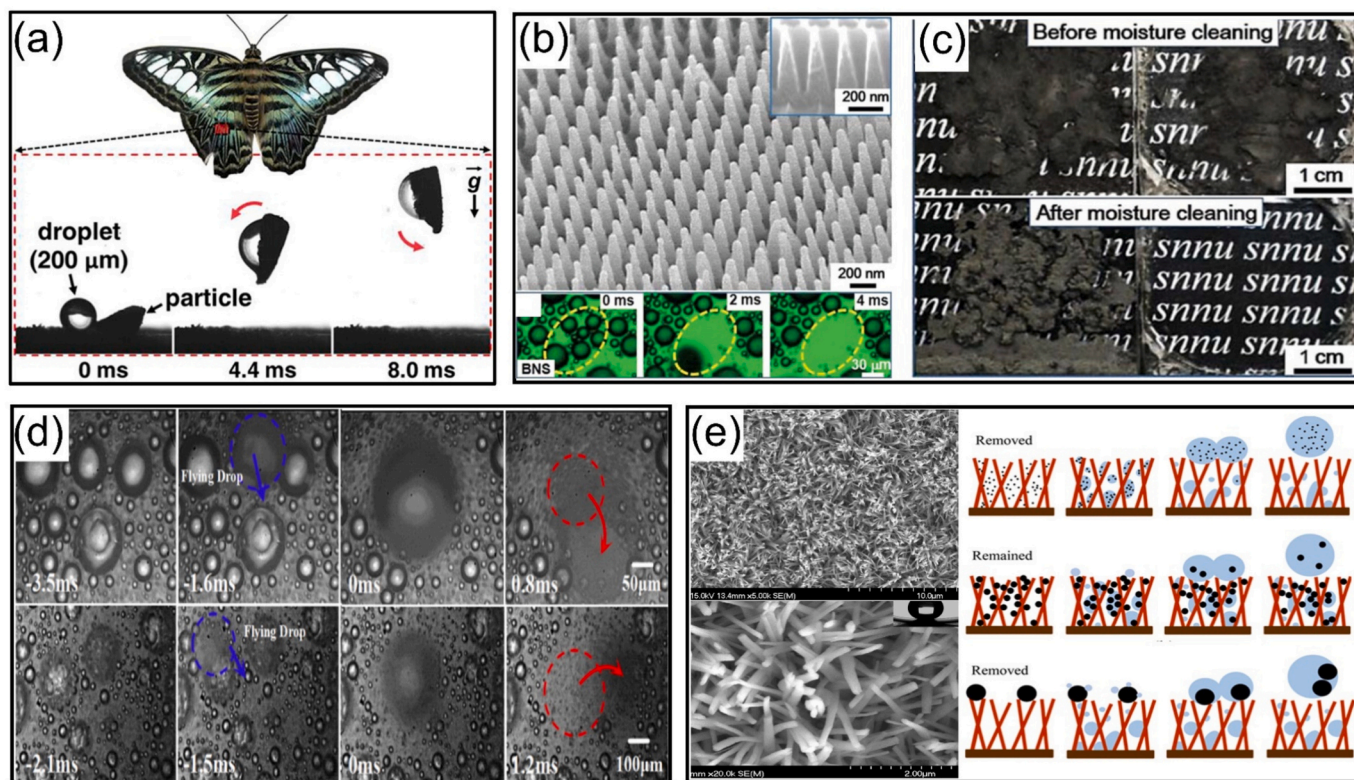


Fig. 13. The applications of droplet jumping in self-cleaning. (a) bionic droplet-particle jumping nanostructured SHS [171]. Copyright 2022 American Chemical Society; (b) bionic nano-cone array self-cleaning surface [173]. Copyright 2019 John Wiley and Sons; (c) optical images of the contaminated bionic surface moisture cleaning [173]. Copyright 2019 John Wiley and Sons; (d) scenarios for autonomous removal of particles via water self-propelled jumping of liquid condensate [174]; Copyright 2018 Elsevier; (e) SEM images of copper surfaces and schematic images of a self-cleaning mechanism [175]. Copyright 2023 American Chemical Society.

maintains an equilibrium temperature above 0 °C. Needle-cluster surfaces with fine gaps and tip structures have demonstrated non-adhesion to supercooled water, effectively preventing ice formation even in extreme conditions [177]. Zhao et al. [178] developed a superhydrophobic aluminum surface with hierarchical pyramidal structures. The coalescence jumping allowed droplets to remain in the Cassie state and detach rapidly, achieving efficient anti-frost performance (Fig. 14b). Other strategies include applying low-surface-energy coatings to promote spontaneous droplet removal, providing a simple and scalable method for anti-icing and anti-frosting [179].

Delaying the growth of ice bridges to reduce the rate of frosting and icing is also a viable approach. According to recent findings, surface frost is closely linked to the spatial distribution of droplets [180]. The total surface's frost depends on the diffusion of ice bridges between individual droplets (Fig. 14c). Supercooled droplets jump, preventing ice bridges from connecting, minimizing chain reactions, preserving the dryness of the immediate surface, and accomplishing anti-frost function. By stimulating supercooled droplets to jump, Chen et al. [181] created a superhydrophobic surface with dual roughness that prevented ice nucleation and ice bridge connection. The surface defrosted quickly, and frost development was delayed. Patterned surfaces have been shown to change the surface's hydrophilicity and hydrophobicity and help prevent frost. Inspired by frosted leaves, Yao et al. [182] investigated the frosting process of serrated surface features and demonstrated pattern design create frost-free zones (Fig. 14d), thereby limiting frost expansion. Recent studies have revealed that the vaporization momentum generated by the interfacial pressure imbalance can trigger frozen droplet jumping, effectively breaking ice bridges and contributing to anti-icing functionality (Fig. 14e) [183,184]. Inspired by frozen droplet jumping driven by vaporization momentum, spring-like pillar surfaces have been designed and fabricated to store elastic energy from the volume expansion of freezing droplets, which is later released as kinetic

energy to drive frozen droplet ejection (Fig. 14f) [185].

6. Summary and outlook

Since the pioneering work of Boreyko and Chen [16], spontaneous droplet jumping has been studied for nearly two decades. We review the two primary mechanisms generated by surface energy and provide a comprehensive summary of the kinetic models, patterns, and influencing factors of droplet spontaneous jumping. Several obstacles and challenges still hinder further research and application.

- I. The key to utilizing the droplet jumping phenomenon is understanding its dynamic characteristics. However, the dynamics of single-droplet jumping remain unclear. For instance, the critical radius of single-droplet jumping in different structures is different and lacks uniform criteria. Non-dimensionalizing the single-droplet jumping velocity using an energy-based approach like coalescence jumping may be feasible. For single-droplet jumping on different structures, selecting a universal characteristic length—such as the effective radius of a confined droplet or a free droplet—may help simplify the problem. Scholars have thoroughly studied the dynamics of equal-sized droplets in multi-droplet jumping. However, research on the more frequently occurring multi-droplet and non-equal-sized droplet jumping is still limited.
- II. Detailed observations of droplet jumping are primarily limited to the millimeter scale. Due to the imaging limitations of the equipment, droplet jumping at the micrometer and nanometer scales can only demonstrate essential characteristics such as speed and direction. At the same time, it is nearly impossible to observe detailed morphological changes. Scaling has become the primary method for dynamic research. However, there is a lack of

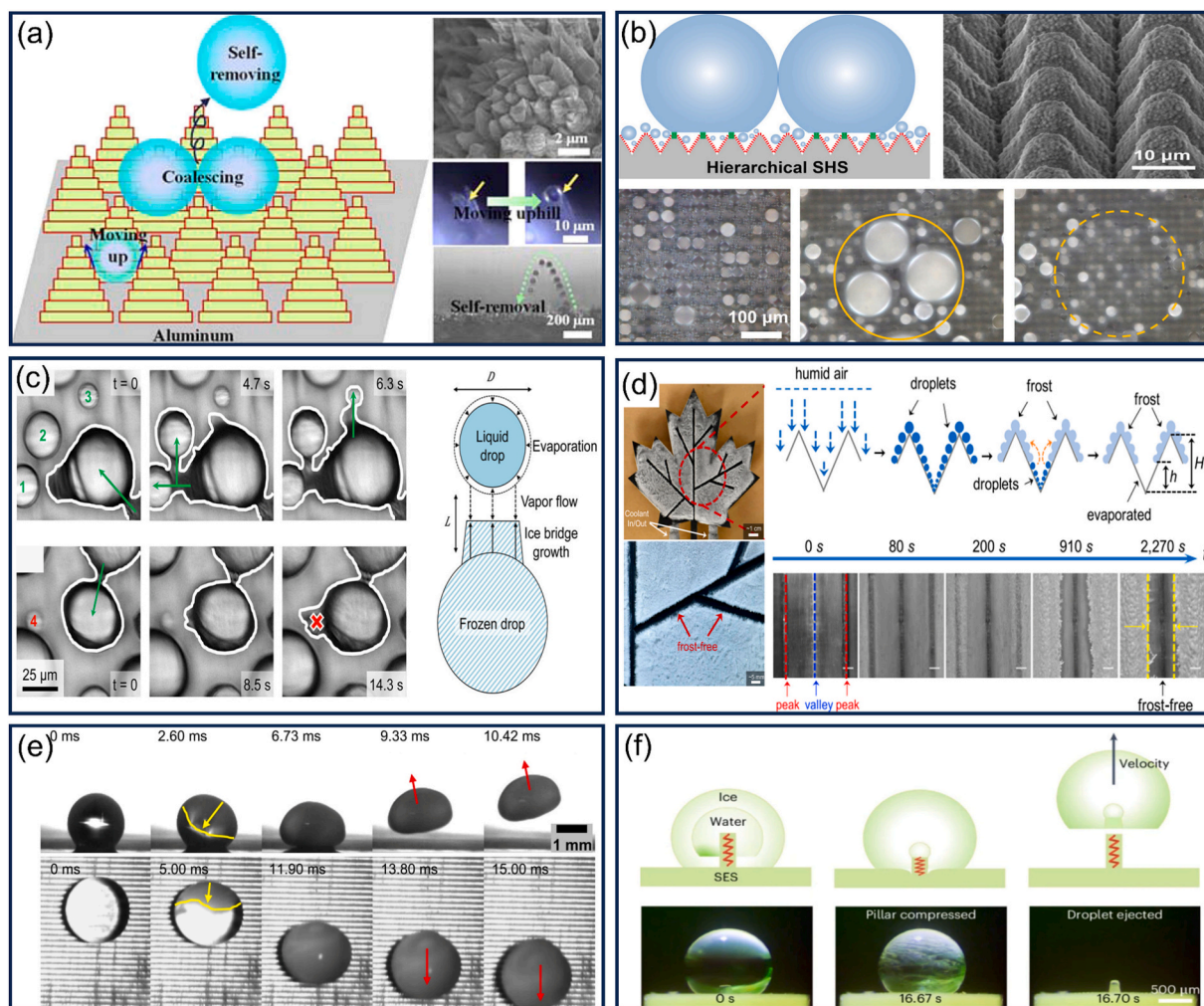


Fig. 14. The applications of droplet jumping in anti-icing and anti-fogging. (a) spontaneous uphill movement and self-removal of condensates [176]. Copyright 2016 American Chemical Society; (b) hierarchical nanostructure droplet jumping anti-frost surface [178]. Copyright 2020 American Chemical Society; (c) ice bridge growth from a frozen drop to neighboring liquid drops on a hydrophobic surface [180]. Copyright 2013 American Chemical Society; (d) anti-frosting patterned surfaces inspired by the veins of leaves. [182] Copyright 2020 National Academy of Sciences; (e) freezing droplet jumping driven by vaporization momentum [184]. Copyright 2024 Springer Nature; (f) spring-pillar surface driving frozen droplet jumping [185]. Copyright 2024 Springer Nature.

more precise experimental results to confirm the accuracy of scaling analyses. More refined experimental methods need to be developed. Scanning electron microscopy (SEM) has been widely used in the study of microdroplet nucleation [186], and its application to droplet jumping may help address observation challenges. In addition, particle image velocimetry (PIV) [187] could be employed to resolve the difficulty of visualizing internal flow during experiments by providing insights into the droplet's flow field.

- III. There is limited research on active methods to promote droplet jumping, and the excitation techniques involving magnetic and electric fields still require detailed investigation. Additionally, there is insufficient research on combining active and passive approaches to enhance droplet jumping. Combining these methods could enhance their benefits while reducing their respective drawbacks. Research on passive methods to improve droplet jumping is more extensive. However, there is a lack of design guidance for specific surface types, such as columnar or ridge-like surfaces. Additionally, studies on directing the movement of droplet jumping are relatively scarce. Machine learning-guided design of structured surfaces may be the key to solving the challenge of multi-parameter structural optimization.

- IV. Numerical simulations can effectively replicate the droplet coalescence-induced jumping phenomenon; however, challenges may arise when simulating single or multiple droplet jumping (three or more). Difficulties also occur when defining dynamic contact angles and managing moving three-phase contact line boundary conditions. Since the nanoscale cannot capture the details of droplet jumping, simulations at this scale still face credibility issues. Compared to experimental methods, numerical simulation techniques have received less research attention. For millimeter-scale droplets, validating multi-droplet simulation methods by existing experimental results is a robust approach. In the case of nanodroplets, molecular dynamics (MD) simulations [188] are more commonly used, and their reliability can be further ensured by comparing the simulated nucleation morphology or contact angle—features that can be directly observed at the nanoscale.
- V. Promoting valuable applications is the most significant barrier to developing droplet-jumping technology. Pure or deionized water makes up most experimental droplets, independent of the water's corrosiveness, pH, salinity, or impurities. There is less research on other liquids commonly used in industry, like oil and refrigerants. Moreover, improving superhydrophobic surfaces' long-term chemical stability and mechanical robustness is a significant

challenge. Most methods for producing superhydrophobic surfaces are complicated, expensive, and time-consuming. Thus, the primary impediment to the widespread application of droplet jumping technology is the creation of durable, superhydrophobic surfaces. We must complete this affordably, efficiently, and promptly. In materials research, designing cost-effective and durable superhydrophobic structures [189] or coatings [190] is essential to addressing this challenge.

CRedit authorship contribution statement

Zijin Zhang: Writing – original draft, Methodology, Investigation, Formal analysis. **Jin Wang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yongqing He:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Feng Jiao:** Writing – review & editing, Supervision.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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Data availability

The data supporting this study's findings are available from the corresponding author upon reasonable request.

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